

Document

Project: DevOps Implementation

Note: Contents have been deleted. This is not a full version of the document.

Date: Sept 2025

Prepared by

Author

Jean-Philippe LO SIOU

Version: 1.0

DOCUMENT CONTROL

Date	Author	Ver	Comments
11/03/2025	Jean-Philippe LO SIOU	0.1	First Drafts
14/03/2025	Jean-Philippe LO SIOU	0.2	First Release
17/03/2025	Jean-Philippe LO SIOU	0.3	Various Minor Updates (Formatting)
18/03/2025	Jean-Philippe LO SIOU	0.4	Adding challenges
29/03/2025	Jean-Philippe LO SIOU	0.5	Adding Challenges
30/03/2025	Jean-Philippe LO SIOU	0.6	Various Updates
02/04/2025	Jean-Philippe LO SIOU	0.7-1.0	Final Release

Contents

Project: DevOps Implementation	1
Document Control	2
Objective.....	4
Expected Deliverables	4
Selection & Forking a Repository.....	5
Git & Version Control.....	5
Containerization with Docker & Kubernetes.....	9
Deployment in Local Docker Container	9
Deployment in Local Kubernetes Clusters.....	18
Implementing Infrastructure as Code (IaC)	29
Terraform.....	29
Deployment in Azure Kubernetes Cluster through Terraform	40
Setting Up CI/CD Pipeline	55
GitHub Actions Workflow Push Docker image to private repository	55
GitHub Actions Workflow Terraform.....	60
Workflow Unit Test.....	60
Workflow Init Backend	64
Workflow Azure Kubernetes Services	65
Workflow Applications to Azure Kubernetes Services	68

OBJECTIVE

This project consists of hands-on experience in DevOps practices by selecting an open-source repository of their interest, implementing automation, and creating a fully integrated CI/CD pipeline. By the end, this project is a fully automated system where changes are automatically tested, deployed, and maintained.

Expected Deliverables

1. A **GitHub repository** containing the forked project with all DevOps implementations.
2. A **fully automated CI/CD pipeline**.
3. An **Infrastructure as Code (IaC) setup**.
4. A **containerized application deployed to the cloud**.
5. A **detailed report explaining the execution, challenges faced, and learnings**.

SELECTION & FORKING A REPOSITORY

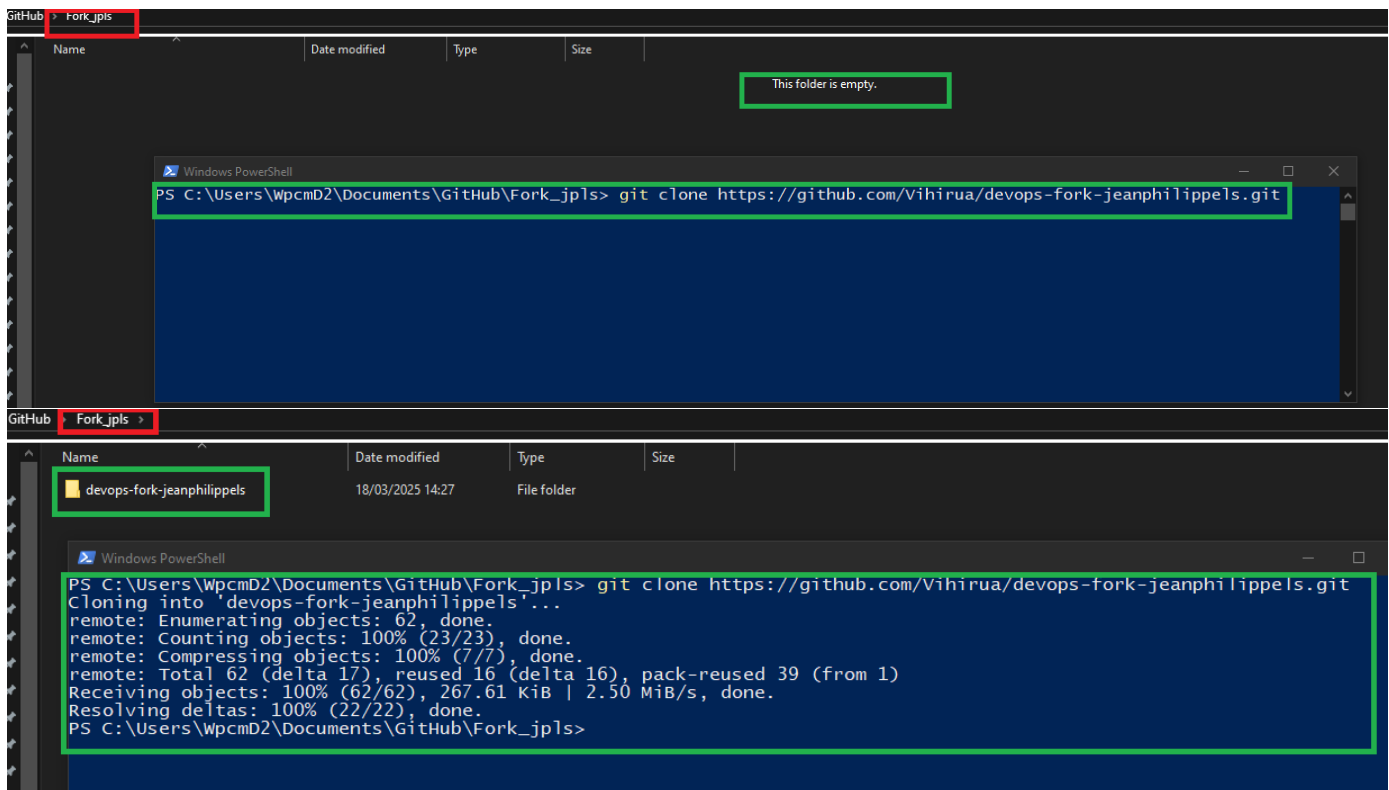
Git & Version Control

- **Forking a project**
- **Create a new branch for DevOps changes**
- **Practice Git Workflow:**
 - **Make changes and commit**
 - **Push changes to GitHub.**
 - **Create a Pull Request (PR) and merge changes.**

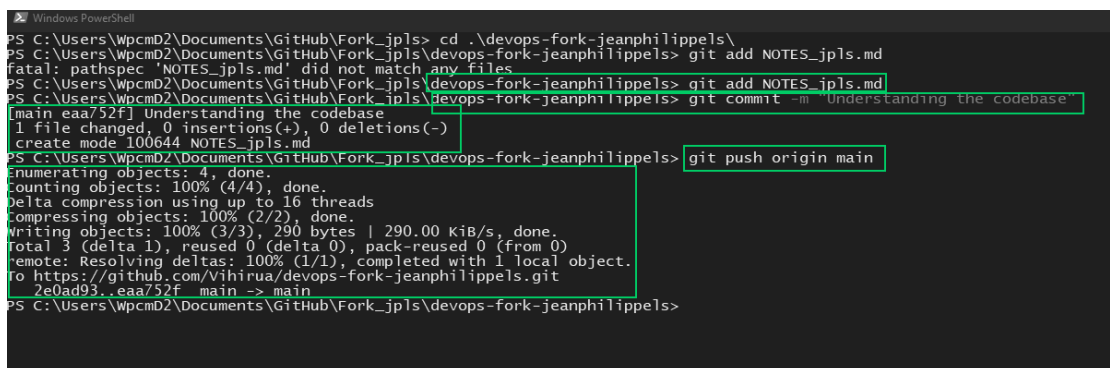
- **Forking a project**

The screenshot shows the GitHub interface for creating a new fork. At the top, the repository path 'felixcameron17 / project-setup-environments' is highlighted with a red box. Below the repository name, a red box contains the text 'Create a new fork'. The form includes fields for 'Owner' (Vihirua) and 'Repository name' (devops-fork-jeanphilippel), which are highlighted with a green box. A green checkmark indicates that the repository name is available. Below these fields, there is a checkbox for 'Copy the main branch only' which is checked. At the bottom right, a green box highlights the 'Create fork' button.

The screenshot shows the GitHub repository page for 'devops-fork-jeanphilippels' by user 'Vihirua'. The repository name is highlighted with a green box. Below the repository name, it says 'forked from felixcameron17/project-setup-environments'. The page shows the 'main' branch with 1 branch and 0 tags. There is a 'Go to file' search bar and a 'Code' button. Below this, there is a section for 'This branch is up to date with felixcameron17/project-setup-environments:main.' with 'Contribute' and 'Sync fork' buttons. A list of files is shown, including .github/workflows, MainPage.js, README.md, banner.jpg, banner2.jpg, bootstrap-grid.min.css, bootstrap-reboot.min.css, and bootstrap.min.css. On the right side, there is a sidebar with user profile information and navigation links.



- **Git clone**
- **Create a new branch for DevOps changes**



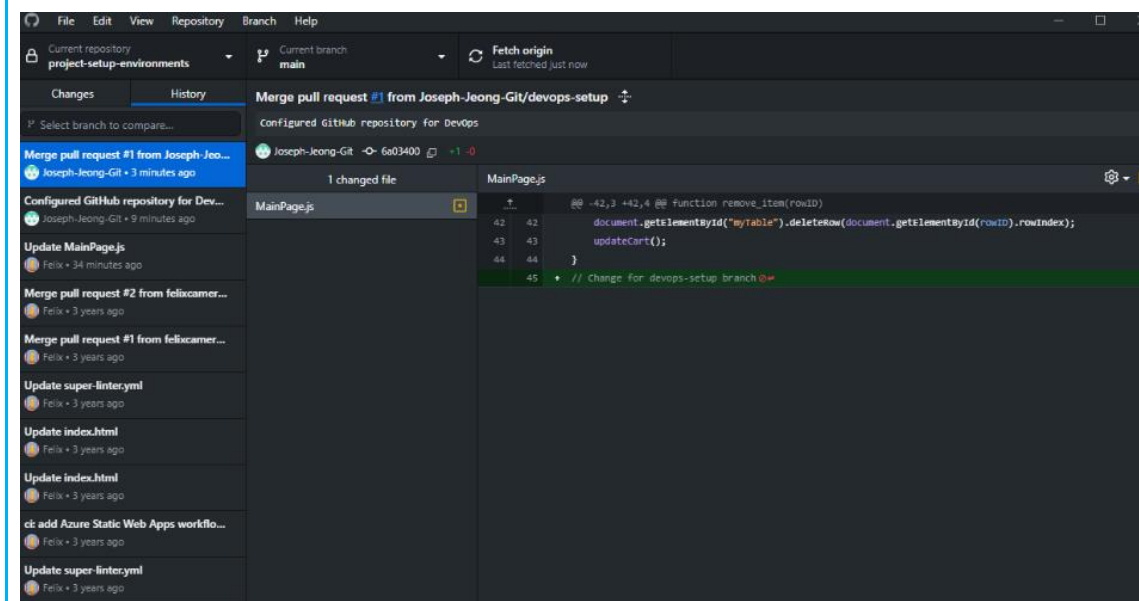
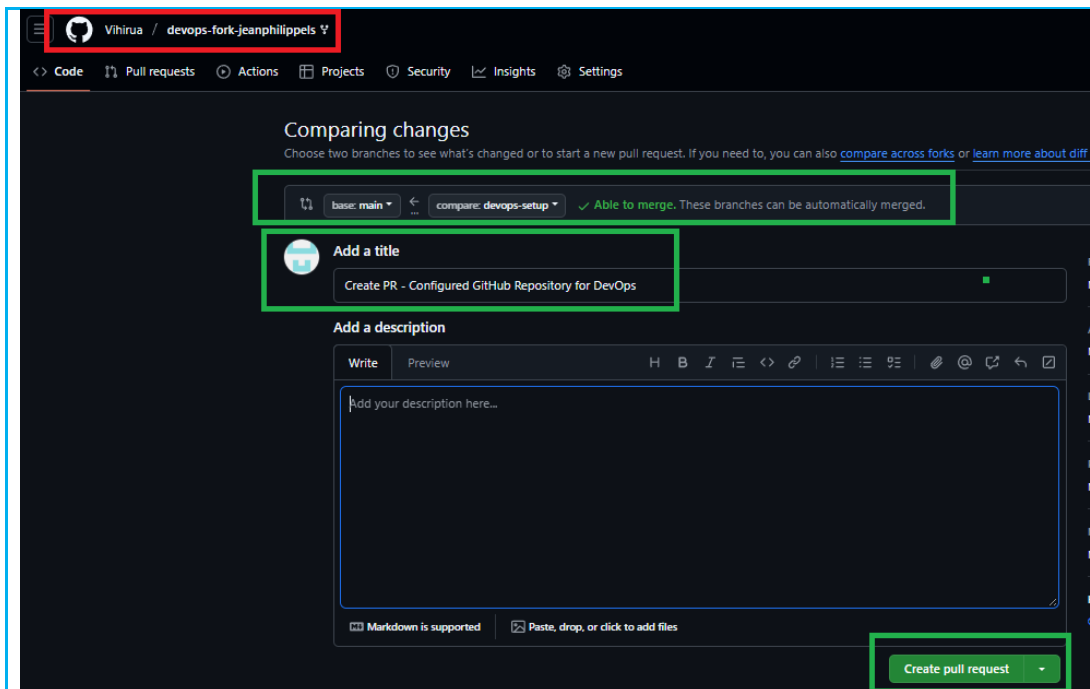
- **Git add NOTES_jpls.md**
- **Git commit -m "Understanding the codebase"**

- **Git push origin main**

- **Push changes to GitHub.**

```
Windows PowerShell
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls> cd .\devops-fork-jeanphilippels\
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels> git add NOTES_jpls.md
fatal: pathspec 'NOTES_jpls.md' did not match any files
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels> git add NOTES_jpls.md
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels> git commit -m "Understanding the codebase"
[main eaa752f] Understanding the codebase
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 NOTES_jpls.md
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels> git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 290 bytes | 290.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Vihirua/devops-fork-jeanphilippels.git
 2e0ad93..eaa752f  main -> main
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels>
```

- **Create a Pull Request (PR) and merge changes.**



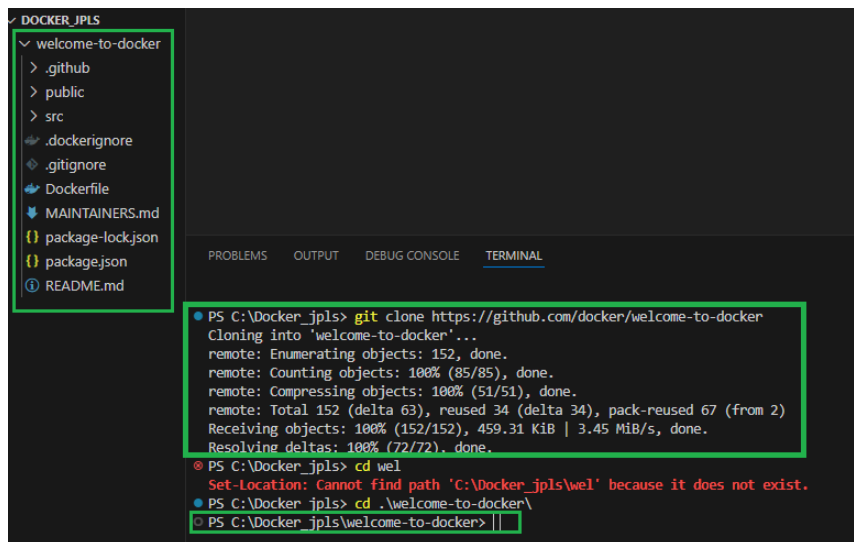
CONTAINERIZATION WITH DOCKER & KUBERNETES

Deployment in Local Docker Container

- **Run a Docker container**
- **Push Docker Image to Docker Hub.**
- **Building DevOps image with Dockerfile**
- **Building container with the DevOps image**
- **Testing container with the DevOps image**
- **Get into the DevOps docker container**
- **Pushing DevOps image to private docker repository**

Docker is a software allowing to build, test and deploy applications by using containers. It packages everything in containers that the applications need to run with (Libraries, systems tools, code). It can quickly deploy and scale applications into any environments (any operating Systems)

- **Run a Docker container**



```
PS C:\Docker_jpls> git clone https://github.com/docker/welcome-to-docker
Cloning into 'welcome-to-docker'...
remote: Enumerating objects: 152, done.
remote: Counting objects: 100% (85/85), done.
remote: Compressing objects: 100% (51/51), done.
remote: Total 152 (delta 63), reused 34 (delta 34), pack-reused 67 (from 2)
Receiving objects: 100% (152/152), 459.31 KiB | 3.45 MiB/s, done.
Resolving deltas: 100% (72/72), done.
PS C:\Docker_jpls> cd wel
Set-Location: Cannot find path 'C:\Docker_jpls\wel' because it does not exist.
PS C:\Docker_jpls> cd .\welcome-to-docker\
PS C:\Docker_jpls\welcome-to-docker> |
```

- **Git clone**

The Dockerfile :

```

1 # Start your image with a node base image
2 FROM node:18-alpine
3
4 # The /app directory should act as the main application directory
5 WORKDIR /app
6
7 # Copy the app package and package-lock.json file
8 COPY package*.json ./
9
10 # Copy local directories to the current local directory of our docker image (/app)
11 COPY ./src ./src
12 COPY ./public ./public
13
14 # Install node packages, install serve, build the app, and remove dependencies at the end
15 RUN npm install \
16     && npm install -g serve \
17     && npm run build \
18     && rm -fr node_modules
19
20 EXPOSE 3000
21
22 # Start the app using serve command
23 CMD [ "serve", "-s", "build" ]

```

```

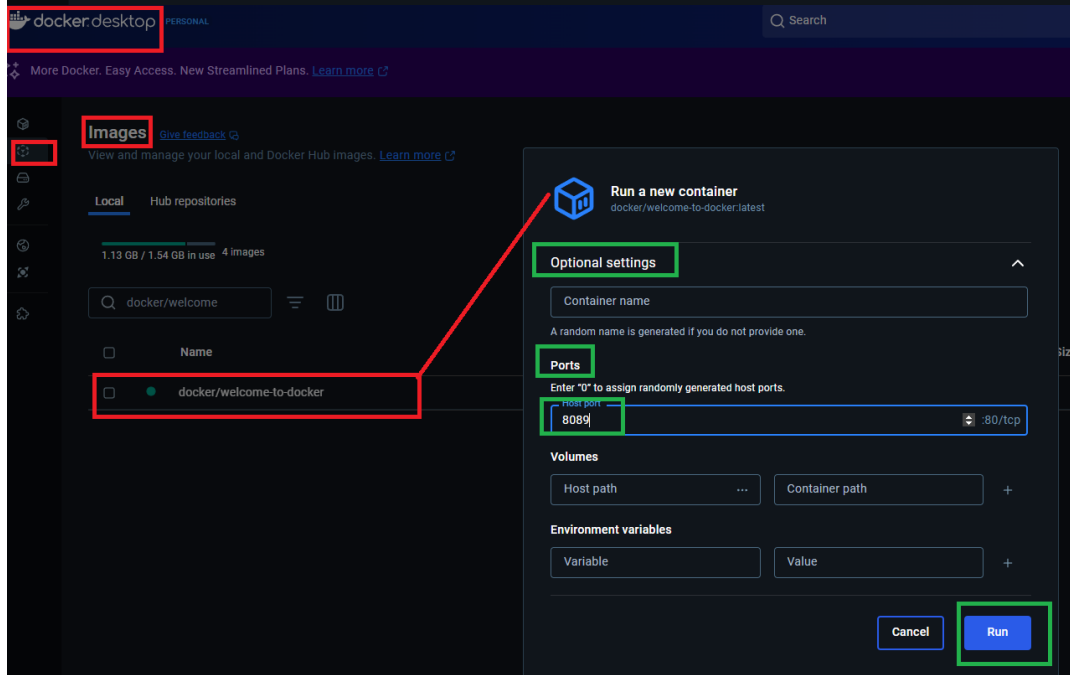
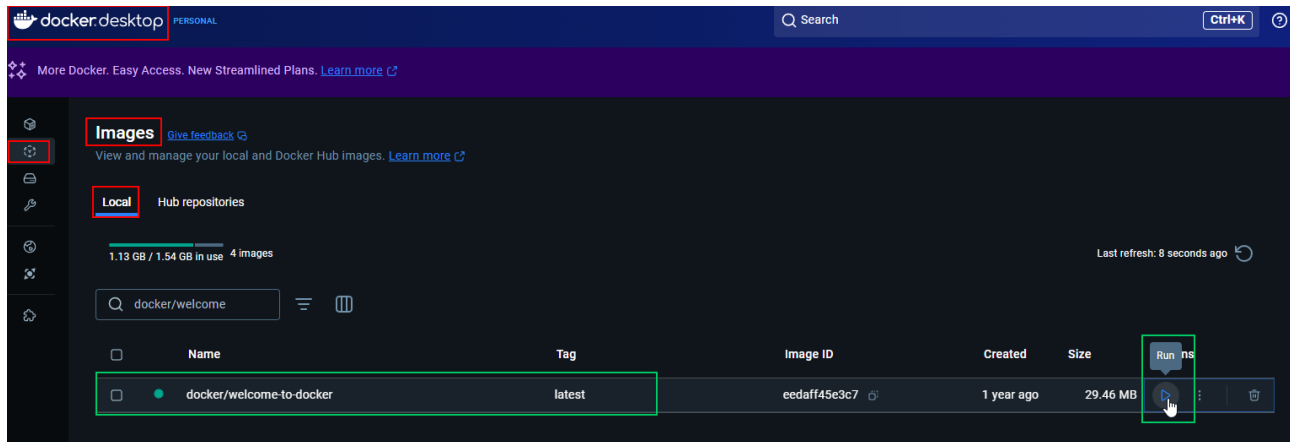
PS C:\Docker_jpls\welcome-to-docker> docker build -t welcome-to-docker .
[*] Building 3.0s (3/3)

-> [internal] load build definition from Dockerfile
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> transferring dockerfile: 647B 0.0s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.0s
-> [auth] library/node:pull token for registry-1.docker.io 0.0s
-> sha256:f46e519824fbb18a1df4becbda40665f5c1cd193e527b3b0e26683d10f140916 1.20MB / 1.20MB 0.5s
-> [1/6] FROM docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 3.0s
-> sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 40.01MB / 40.01MB 1.1s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.3s
-> extracting sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 0.8s
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> extracting sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.0s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.0s
-> extracting sha256:f46e519824fbb18a1df4becbda40665f5c1cd193e527b3b0e26683d10f140916 1.20MB / 1.20MB 0.1s
-> [4/6] COPY ./src ./src 0.1s
-> sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 40.01MB / 40.01MB 0.1s
-> transferring context: 691.39kB 0.0s
-> extracting sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 0.0s
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> [5/6] COPY package*.json ./ 0.1s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.0s
-> sha256:f46e519824fbb18a1df4becbda40665f5c1cd193e527b3b0e26683d10f140916 1.20MB / 1.20MB 0.1s
-> [internal] load build context
-> transferring context: 52B 0.0s
[*] Building 52.5s (12/12) FINISHED

-> [internal] load build definition from Dockerfile
-> transferring dockerfile: 647B 0.0s
-> [internal] load metadata for docker.io/library/node:18-alpine 2.9s
-> [auth] library/node:pull token for registry-1.docker.io 0.0s
-> [internal] load .dockerignore 0.0s
-> resolving paths 0.0s
-> [1/6] FROM docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.3s
-> sha256:f46e519824fbb18a1df4becbda40665f5c1cd193e527b3b0e26683d10f140916 1.20MB / 1.20MB 0.1s
-> sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 40.01MB / 40.01MB 0.8s
-> extracting sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 0.1s
-> extracting sha256:f46e519824fbb18a1df4becbda40665f5c1cd193e527b3b0e26683d10f140916 0.1s
-> extracting sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 0.0s
-> [internal] load build context
-> transferring context: 691.39kB 0.0s
-> [2/6] WORKDIR /app 0.0s
-> [3/6] COPY package*.json ./ 0.1s
-> [4/6] COPY ./src ./src 0.1s
-> [5/6] COPY ./public ./public 0.1s
-> [6/6] RUN npm install && npm install -g serve && npm run build && rm -fr node_modules 39.0s
-> exporting image 6.8s
-> exporting layers 4.8s
-> exporting manifest sha256:c36647e66994f7f246b3f85fb7aad7980bfdd158c40e5262b4646b6abc5f 0.0s
-> exporting config sha256:63979c85110897d40e4048a0e571c727dad8a6f996cc3a0570ff80c3706 0.0s
-> exporting attestation manifest sha256:704d99747f282f80d56a74bdeb27644332326e2d0970fc4818bfd215044 0.1s
-> exporting manifest list sha256:53c058ff61fa7a4f9203839c0e0dbcc1e4f996cd9f0219f13a99ced094293e 0.0s
-> naming to docker.io/library/welcome-to-docker:latest 0.0s
-> unpacking to docker.io/library/welcome-to-docker:latest 1.7s

```

- `Docker build -t welcome-to-docker .`



The screenshot displays the Docker Desktop interface. At the top, the 'Containers' tab is selected, showing a container named 'nifty_noether' with ID 'c428c7f979bb' and port '8089:80'. The 'Logs' tab is active, showing a series of log messages from the container's entrypoint script, including configuration steps and the start of various worker processes. Below the logs, a 'Welcome to Docker' window is open, displaying a 'Congratulations!!!' message and the text 'You ran your first container.' with social media sharing buttons for X, LinkedIn, and Docker.

- Push Docker Image to Docker Hub.

```
PS C:\Docker_jpls\welcome-to-docker> docker images --filter=reference=welcome-to-docker
REPOSITORY          TAG         IMAGE ID      CREATED       SIZE
welcome-to-docker    latest      53c050ff61fa  23 minutes ago  383MB
PS C:\Docker_jpls\welcome-to-docker> docker tag welcome-to-docker dockerlsp123/welcome-to-docker-jpls:latest
```

- Docker images
- Docker tag welcome-to-docker dockerlsp123/welcome-to-docker-jpls:latest

```
PS C:\Docker_jpls\welcome-to-docker> docker images --filter=reference=dockerlsp123/welcome-to-docker-jpls
REPOSITORY          TAG         IMAGE ID      CREATED       SIZE
dockerlsp123/welcome-to-docker-jpls  latest      53c050ff61fa  24 minutes ago  383MB
PS C:\Docker_jpls\welcome-to-docker>
```

```
PS C:\Docker_jpls\welcome-to-docker> docker login
Authenticating with existing credentials...
Login Succeeded
```

```
PS C:\Docker_jpls\welcome-to-docker> docker push dockerlsp123/welcome-to-docker-jpls:latest
The push refers to repository [docker.io/dockerlsp123/welcome-to-docker-jpls]
f18232174bc9: Mounted from dockerlsp123/devops-jeanphilippels
f46e519824fb: Pushing [=====>] 1.261MB/1.261MB
f7f4b521746b: Pushed
31e2774c5eb0: Pushing [====>] 4.194MB/62.88MB
3d8c59f7308d: Pushed
a5cf150cb374: Pushing [=====>] 6.291MB/40.01MB
6e0bb1434e7f: Pushed
e6f3dfa2eb18: Pushed
e28bbca3bfc7: Pushed
f9ec8daafabc: Pushed

```

- Docker push dockerlsp123/welcome-to-docker-jpls:latest

```
PS C:\Docker_jpls\welcome-to-docker> docker push dockerlsp123/welcome-to-docker-jpls:latest
The push refers to repository [docker.io/dockerlsp123/welcome-to-docker-jpls]
f18232174bc9: Mounted from dockerlsp123/devops-jeanphilippels
f46e519824fb: Pushed
f7f4b521746b: Pushed
31e2774c5eb0: Pushed
3d8c59f7308d: Pushed
a5cf150cb374: Pushed
6e0bb1434e7f: Pushed
e6f3dfa2eb18: Pushed
e28bbca3bfc7: Pushed
f9ec8daafabc: Pushed
latest: digest: sha256:53c050ff61fa7a4f9203839c60e8cbcc1e4f996cd9f6219f33ad99ced694293e size: 856
PS C:\Docker_jpls\welcome-to-docker>
```

Repositories / welcome-to-docker-jpls / General

dockerlsp123/welcome-to-docker-jpls

Last pushed 2 minutes ago · Repository size: 103 MB

Add a description

Add a category

General Tags Image Management **BETA** Collaborators Webhooks Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	2 minutes

[See all](#)

- Building DevOps image with Dockerfile

```
1 FROM nginx:alpine
2
3 COPY ./src /usr/share/nginx/html
4
5 EXPOSE 80
```

```
PS C:\Docker_jpls> docker login
Authenticating with existing credentials...
Login did not succeed, error: error during connect: Post "http://%2f%2F.%2FdockerDesktopLinuxEngine/v1.47/auth": open //.pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

USING WEB-BASED LOGIN
To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: VPMH-NZHQ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
█
```



Device Confirmation

Please confirm this is the code displayed on your Docker CLI:

VMHW-NZHQ

If you did not initiate this action or you do not recognize this device select cancel.

Cancel

Confirm

```
PS C:\Docker_jpls> docker build -t devops-apps-jeanphilippels .
[*] Building 6.2s (8/8) FINISHED
-> [internal] load build definition from Dockerfile
-> transferring dockerfile: 103B
-> [internal] load metadata for docker.io/library/nginx:alpine
-> [auth] library/nginx:pull token for registry-1.docker.io
-> [internal] load .dockerignore
-> transferring context: 2B
-> [internal] load build context
-> transferring context: 547.46kB
-> [1/2] FROM docker.io/library/nginx:alpine@sha256:4ff102c5d78d25a6f0da662b3cf39eaf07f01e0c0927fd21e219d0af8bc0591
-> resolve docker.io/library/nginx:alpine@sha256:4ff102c5d78d25a6f0da662b3cf39eaf07f01e0c0927fd21e219d0af8bc0591
-> sha256:6d79c6084d434876ce0f83c675d20532f28e476283a29e7e63bc0bf134ed6 1.40kB / 1.40kB
-> sha256:0c7e4c092ab716c95e1a8840943549b4ac5e6c5d1acc74496528e1b03e1e67a 15.38kB / 15.38kB
-> sha256:ab3286a7346303a31b09a5189f63f1414c1de44e397088dc07edb322df1fe9 1.21kB / 1.21kB
-> sha256:8d27c072a58f81ecf2425172ac8e5b25010ff2d014f80de35090104e462568eb 402B / 402B
-> sha256:984583bcf083fa6900b5e7834795a9a57a9b4dfe7448d5350474f5d309625ece 955B / 955B
-> sha256:43f2ec460bdf5abae56d7e13b060b308f42cbf2103f6665c349f8698dac3962 627B / 627B
-> sha256:ccc35e35d420d0dd115290f1074f6cad1c474b6f94897434e6befd33be781d2 1.79MB / 1.79MB
-> sha256:f18232174bc91741fd3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB
-> extracting sha256:f18232174bc91741fd3da96d85011092101a032a93a388b79e99e69c2d5c870
-> extracting sha256:ccc35e35d420d0dd115290f1074f6cad1c474b6f94897434e6befd33be781d2
-> extracting sha256:43f2ec460bdf5abae56d7e13b060b308f42cbf2103f6665c349f8698dac3962
-> extracting sha256:984583bcf083fa6900b5e7834795a9a57a9b4dfe7448d5350474f5d309625ece
-> extracting sha256:8d27c072a58f81ecf2425172ac8e5b25010ff2d014f80de35090104e462568eb
-> extracting sha256:ab3286a7346303a31b09a5189f63f1414c1de44e397088dc07edb322df1fe9
-> extracting sha256:6d79c6084d434876ce0f83c675d20532f28e476283a29e7e63bc0bf134ed6
-> extracting sha256:0c7e4c092ab716c95e1a8840943549b4ac5e6c5d1acc74496528e1b03e1e67a
-> [2/2] COPY ./src /usr/share/nginx/html
-> exporting to image
-> exporting layers
-> exporting manifest sha256:ea324e542456625fc075c95f664b929071b8d1f3e5cb33216a13a3e2dcfc41d4
-> exporting config sha256:716dc1bc10521e745bac7af0c0a800b0c0fda12c78c1cebd67beb1ddd79a8a65
-> exporting attestation manifest sha256:e3d0aded6e9310d0e40f486f76ba36b514c87ae4f346c9104c3e489305eb658
-> exporting manifest list sha256:1b5acc1844ee7b2bd4cac941f542828e4019cc833c84ee59ac9326f351b09e2d
-> naming to docker.io/library/devops-apps-jeanphilippels:latest
-> unpacking to docker.io/library/devops-apps-jeanphilippels:latest
```

- `docker build -t devops-apps-jeanphilippels .`

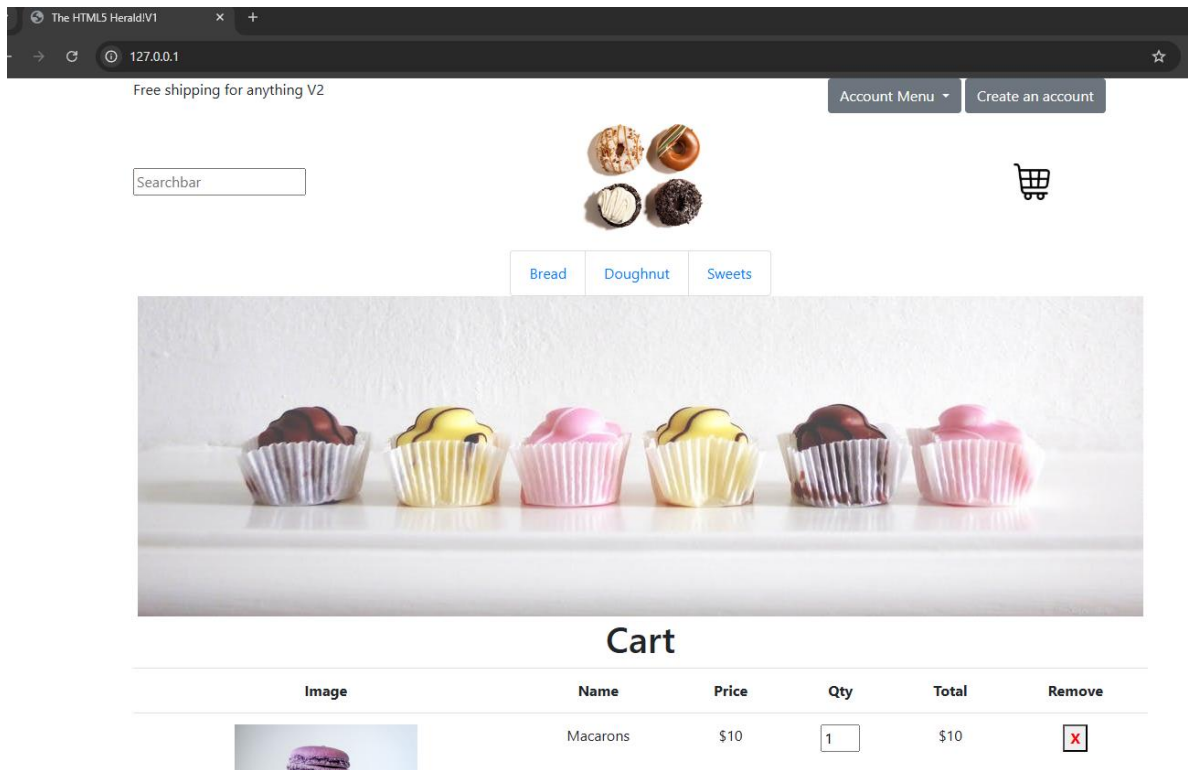
```
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
devops-apps-jeanphilippels latest       1b5acc1844ee     18 seconds ago  74MB
kindest/node         <none>       f226345927d7     3 weeks ago     1.49GB
PS C:\Docker_jpls>
```

- `docker images`
- Building container with the DevOps image

```
PS C:\Docker_jpls> docker run -d -p 80:80 --name devops-jeanphilippels-container devops-apps-jeanphilippels:latest
381164ffdc150bc0fed607615b8172fb8af38a20c03084e32c9336b452399d0c
PS C:\Docker_jpls> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS          PORTS          NAMES
381164ffdc15   devops-apps-jeanphilippels:latest   "/docker-entrypoint..." 5 seconds ago    Up 5 seconds    0.0.0.0:80->80/tcp   devops-jeanphilippels-container
PS C:\Docker_jpls>
```

- `docker run -d -p 80:80 --name devops-jeanphilippels-container devops-apps-jeanphilippels:latest`
- `docker ps`

- Testing container with the DevOps image



- Get into the DevOps docker container

```
PS C:\Docker_jpls> docker exec -it devops-jeanphilippels-container sh
/ # ls -la
total 80
drwxr-xr-x 1 root root 4096 Mar 14 03:33 .
drwxr-xr-x 1 root root 4096 Mar 14 03:33 ..
-rwxr-xr-x 1 root root 0 Mar 14 03:33 .dockerenv
drwxr-xr-x 2 root root 4096 Feb 13 23:04 bin
drwxr-xr-x 5 root root 340 Mar 14 03:33 dev
drwxr-xr-x 1 root root 4096 Feb 14 19:16 docker-entrypoint.d
-rwxr-xr-x 1 root root 1620 Feb 14 19:16 docker-entrypoint.sh
drwxr-xr-x 1 root root 4096 Mar 14 03:33 etc
drwxr-xr-x 2 root root 4096 Feb 13 23:04 home
drwxr-xr-x 1 root root 4096 Feb 13 23:04 lib
drwxr-xr-x 5 root root 4096 Feb 13 23:04 media
drwxr-xr-x 2 root root 4096 Feb 13 23:04 mnt
drwxr-xr-x 2 root root 4096 Feb 13 23:04 opt
dr-xr-xr-x 298 root root 0 Mar 14 03:33 proc
drwx----- 1 root root 4096 Mar 14 03:36 root
drwxr-xr-x 1 root root 4096 Mar 14 03:33 run
drwxr-xr-x 2 root root 4096 Feb 13 23:04/sbin
drwxr-xr-x 2 root root 4096 Feb 13 23:04/srv
dr-xr-xr-x 11 root root 0 Mar 14 03:33/sys
drwxrwxrwt 2 root root 4096 Feb 13 23:04/tmp
drwxr-xr-x 1 root root 4096 Feb 13 23:04/usr
drwxr-xr-x 1 root root 4096 Feb 13 23:04/var
```


- `docker exec -it devops-apps-jeanphilippels:latest sh`

- Pushing DevOps image to private docker repository

```
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID      CREATED        SIZE
devops-apps-jeanphilippels  latest      1b5acc1844ee  About a minute ago  74MB
kindest/node         <none>      f226345927d7  3 weeks ago   1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID      CREATED        SIZE
dockerlspf123/devops-jeanphilippels  manual1      1b5acc1844ee  3 minutes ago  74MB
devops-apps-jeanphilippels          latest      1b5acc1844ee  3 minutes ago  74MB
kindest/node                        <none>      f226345927d7  3 weeks ago   1.49GB
PS C:\Docker_jpls> []
```

- `docker images`

```
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID      CREATED        SIZE
devops-apps-jeanphilippels  latest      1b5acc1844ee  About a minute ago  74MB
kindest/node         <none>      f226345927d7  3 weeks ago   1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID      CREATED        SIZE
dockerlspf123/devops-jeanphilippels  manual1      1b5acc1844ee  3 minutes ago  74MB
devops-apps-jeanphilippels          latest      1b5acc1844ee  3 minutes ago  74MB
kindest/node                        <none>      f226345927d7  3 weeks ago   1.49GB
PS C:\Docker_jpls> []
```

- `docker tags devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1`
- `Docker images`

```

PS C:\Docker_jpls> docker images
REPOSITORY          TAG             IMAGE ID         CREATED          SIZE
devops-apps-jeanphilippels  latest         1b5acc1844ee    About a minute ago  74MB
kindest/node          <none>         f226345927d7    3 weeks ago      1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlsp123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY          TAG             IMAGE ID         CREATED          SIZE
dockerlsp123/devops-jeanphilippels  manual1        1b5acc1844ee    3 minutes ago    74MB
devops-apps-jeanphilippels          latest         1b5acc1844ee    3 minutes ago    74MB
kindest/node                        <none>         f226345927d7    3 weeks ago      1.49GB
PS C:\Docker_jpls> docker push dockerlsp123/devops-jeanphilippels:manual1
The push refers to repository [docker.io/dockerlsp123/devops-jeanphilippels]
ab3286a73463: Layer already exists
8d27c072a58f: Layer already exists
f18232174bc9: Layer already exists
43f2ec460bdf: Layer already exists
6d79cc6084d4: Layer already exists
98bcf5d5f13a: Pushed
01637544f41d: Pushed
984583bcf083: Layer already exists
ccc35e35d420: Layer already exists
0c7e4c092ab7: Layer already exists
manual1: digest: sha256:1b5acc1844ee7b2bd4cac941f542828e4019cc033c04ee59ac9326f351b89e2d size: 856
PS C:\Docker_jpls>

```

- `docker push dockerlsp123/devops-jeanphilippels:manual1`

The screenshot shows the Docker Hub web interface. The breadcrumb navigation at the top reads 'dockerlsp123 / Repositories / devops-jeanphilippels / Tags'. The repository name 'dockerlsp123/devops-jeanphilippels' is highlighted with a red box. Below the repository name, the 'Tags' tab is selected and highlighted with a red box. A table of tags is displayed, with the 'manual1' tag highlighted by a green box. The table includes columns for TAG, Digest, OS/ARCH, Last pull, and Compressed size. The 'manual1' tag has a digest of 'ea324e542456' and a size of '20.17 MB'. To the right of the tag table, there is a 'docker pull' command snippet and a 'Copy' button.

Deployment in Local Kubernetes Clusters

- **Deploy the container using Kubernetes / Apply Kubernetes Configuration**
- *Configure cluster*
- *Create a namespace*
- *Create a nginx deployment*
- *Create a service nginx nodeport*
- *Create a scale of the nginx deployment*

- Deploy the container using Kubernetes / Apply Kubernetes Configuration
- *Create the custom namespace for devops applications*
 - *Create the Secrets allowing to use private docker repository*
 - *Create the service account*
 - *Create the role cluster for the service account*
 - *Create the pod's role for the service account*
- *Bind the role cluster and pod's role to the service account*
- *Create the custom devops web application deployment with secret from a docker hub repository*
- *Create the service custom devops web application deployment*
- *Testing the devops web deployment*

Kind (Kubernetes IN Docker) is a tool allowing to setup a local Kubernetes clusters using Docker Containers. Local development can be explored by using this tool.

Kubernetes is a container orchestration platform allows to automates deployment, the scaling, and the management of containerized applications

- **Configure cluster**

```
PS C:\kubernetes> kind create cluster --config .\kind.yaml --name cka-cluster-jeanphilippels
Creating cluster "cka-cluster-jeanphilippels" ...
  • Ensuring node image (kindest/node:v1.32.2) ...
  ✓ Ensuring node image (kindest/node:v1.32.2) ...
  • Preparing nodes ...
  ✓ Preparing nodes ...
  • Writing configuration ...
  ✓ Writing configuration ...
  • Starting control-plane ...
  ✓ Starting control-plane ...
  • Installing CNI ...
  ✓ Installing CNI ...
  • Installing StorageClass ...
  ✓ Installing StorageClass ...
  • Joining worker nodes ...
  ✓ Joining worker nodes ...

Set kubectl context to "kind-cka-cluster-jeanphilippels"
You can now use your cluster with:

kubectl cluster-info --context kind-cka-cluster-jeanphilippels

Have a nice day! 🍀
```

```
! kind.yaml
1  # three node (two workers) cluster config
2  kind: Cluster
3  apiVersion: kind.x-k8s.io/v1alpha4
4  nodes:
5  - role: control-plane
6    extraPortMappings:
7      - containerPort: 30001
8        hostPort: 30001
9  - role: worker
10 - role: worker
```

```
Kind create cluster -config .\kind.yaml -name cka-cluster-jeanphilippels
```

```
PS C:\kubernetes> kind get clusters
cka-cluster-jeanphilippels
PS C:\kubernetes>
```

- **Kubectl get clusters**

```
PS C:\kubernetes> kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
cka-cluster-jeanphilippels-control-plane	Ready	control-plane	58m	v1.32.2
cka-cluster-jeanphilippels-worker	Ready	<none>	58m	v1.32.2
cka-cluster-jeanphilippels-worker2	Ready	<none>	58m	v1.32.2

```
PS C:\kubernetes>
```

- **Create a namespace**

Kubernetes namespaces allow to organize partition and organize resources within a Kubernetes cluster. They separate in logical groups of resources within a cluster, making cluster management more efficient. They can denote multiple environments like dev, staging, or production, or for grouping resources that make up an application.

- **Create a nginx deployment**

```
PS C:\kubernetes> kubectl create namespace nginx-apps-jeanphilippels
namespace/nginx-apps-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\nginx-deploy.yml
deployment.apps/nginx-deploy-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\nginx-nodeport.yml
service/nginx-nodeport-jeanphilippels created
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deploy-jeanphilippels	3/3	3	3	23s

```
PS C:\kubernetes>
```

```
PS C:\kubernetes> kubectl get namespace -n nginx-apps-jeanphilippels
```

NAME	STATUS	AGE
default	Active	5h26m
dev-apps-jeanphilippels	Active	5h20m
kube-node-lease	Active	5h26m
kube-public	Active	5h26m
kube-system	Active	5h26m
local-path-storage	Active	5h26m
nginx-apps-jeanphilippels	Active	4h8m

```
PS C:\kubernetes> |
```

- Kubectl create namespace nginx-apps-jeanphilippels
- Kubectl apply -f .\nginx-deploy.yml
- Kubectl apply -f .\nginx-nodeport.yml
- Kubectl get deploy -n nginx-apps-jeanphilippels

```
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deploy-jeanphilippels-5f6bd4879d-4cftr	1/1	Running	0	50s
nginx-deploy-jeanphilippels-5f6bd4879d-gxwqn	1/1	Running	0	50s
nginx-deploy-jeanphilippels-5f6bd4879d-mwvss	1/1	Running	0	50s

```
PS C:\kubernetes> |
```

- Kubectl get pods -n nginx-apps-jeanphilippels

```
PS C:\kubernetes> kubectl edit deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
```

- Example of increase replicas with “edit”

- Kubectl edit deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels

The image displays two sequential screenshots of a Notepad window editing a Kubernetes Deployment YAML file. The window title is 'kubectl.exe-edit-3503174900.yaml - Notepad'. The file content is a YAML configuration for a Deployment.

First Screenshot (Top): The 'replicas' field is set to 3 and is highlighted with an orange box. Other fields include 'apiVersion: apps/v1', 'kind: Deployment', 'name: nginx-deploy-jeanphilippels', 'namespace: nginx-apps-jeanphilippels', and 'resourceVersion: "8152"'. The 'spec' section includes 'progressDeadlineSeconds: 600', 'revisionHistoryLimit: 10', and a 'strategy' with 'rollingUpdate' settings.

Second Screenshot (Bottom): The 'replicas' field has been updated to 5 and is now highlighted with a green box. All other fields remain unchanged from the first screenshot.

```
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deploy-jeanphilippels        5/5    5           5          4m40s
PS C:\kubernetes>
```

- **Kubectl get deploy -n nginx-apps-jeanphilippels**

```
PS C:\kubernetes> kubectl describe deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
Name:                               nginx-deploy-jeanphilippels
Namespace:                           nginx-apps-jeanphilippels
CreationTimestamp:                   Tue, 11 Mar 2025 11:16:49 +1300
Labels:                               env=demo
Annotations:                         deployment.kubernetes.io/revision: 1
Selector:                             env=demo
Replicas:                            5 desired | 5 updated | 5 total | 5 available | 0 unavailable
StrategyType:                        RollingUpdate
MinReadySeconds:                     0
RollingUpdateStrategy:               25% max unavailable, 25% max surge
Pod Template:
  Labels:                             env=demo
  Containers:
    nginx:
      Image:                            nginx
      Port:                             <none>
      Host Port:                         <none>
      Environment:                       <none>
      Mounts:                             <none>
      Volumes:                             <none>
      Node-Selectors:                     <none>
      Tolerations:                         <none>
Conditions:
  Type           Status  Reason
  ----           -
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets: <none>
NewReplicaSet:  nginx-deploy-jeanphilippels-5f6bd4879d (5/5 replicas created)
Events:
  Type           Reason              Age             From              Message
  ----           -
  Normal         ScalingReplicaSet   6m3s            deployment-controller Scaled up replica set nginx-deploy-jeanphilippels-5f6bd4879d from 0 to 3
  Normal         ScalingReplicaSet   92s             deployment-controller Scaled up replica set nginx-deploy-jeanphilippels-5f6bd4879d from 3 to 5
PS C:\kubernetes>
```

- **Kubectl describe deploy/nginx-deploy-jeanphilippels**

- **Example of increase replicas with “scale”**

```
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deploy-jeanphilippels        5/5    5           5          3h33m
PS C:\kubernetes> kubectl scale deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels --replicas=10
deployment.apps/nginx-deploy-jeanphilippels scaled
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deploy-jeanphilippels        7/10   10          7          3h34m
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deploy-jeanphilippels        10/10  10          10         3h34m
```

- **Kubectl get deploy -n nginx-apps-jeanphilippels**
- **Kubectl scale deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels --replicas=10**

- **Kubect1 get deploy -n nginx-apps-jeanphilippels**

```
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deploy-jeanphilippels-5f6bd4879d-4cftr  1/1     Running   1 (99s ago)  3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-4znh4  1/1     Running   0           58s
nginx-deploy-jeanphilippels-5f6bd4879d-7rv42  1/1     Running   0           58s
nginx-deploy-jeanphilippels-5f6bd4879d-g6x8l  1/1     Running   0           58s
nginx-deploy-jeanphilippels-5f6bd4879d-gxwqn  1/1     Running   1 (99s ago)  3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-lqldx  1/1     Running   1 (99s ago)  3h30m
nginx-deploy-jeanphilippels-5f6bd4879d-mwvss  1/1     Running   1 (99s ago)  3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-sndnz  1/1     Running   1 (99s ago)  3h30m
nginx-deploy-jeanphilippels-5f6bd4879d-t6rzt  1/1     Running   0           58s
nginx-deploy-jeanphilippels-5f6bd4879d-vt8cz  1/1     Running   0           58s
PS C:\kubernetes>
```

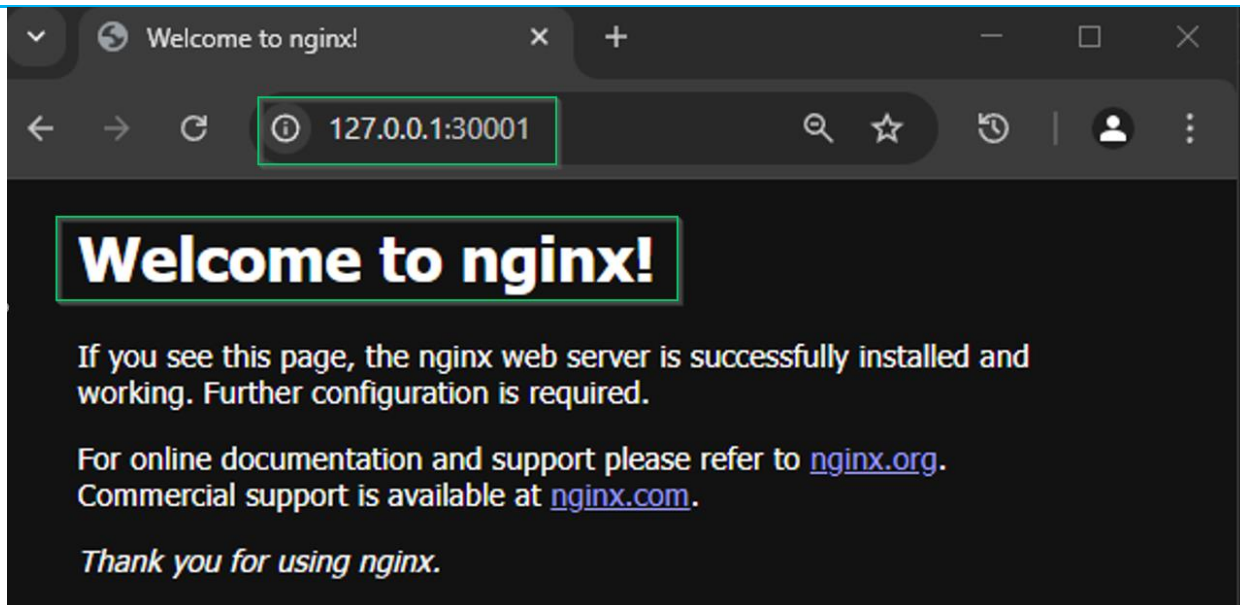
- **Kubect1 get pods -n nginx-apps-jeanphilippels**

```
PS C:\kubernetes> kubectl get svc -n nginx-apps-jeanphilippels
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
nginx-nodeport-jeanphilippels      NodePort    10.96.81.182  <none>       80:30001/TCP    4h5m
PS C:\kubernetes>
```

- **Delete a nginx deployment**

```
PS C:\kubernetes> kubectl delete deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
deployment.apps "nginx-deploy-jeanphilippels" deleted
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
No resources found in nginx-apps-jeanphilippels namespace.
PS C:\kubernetes>
```

- **Kubect1 delete deploy/nginx-deploy-jeanphilippes -n nginx-apps-jeanphilippels**



- Create the custom namespace for devops apps

```
PS C:\kubernetes> kubectl create namespace dev-apps-jeanphilippels
namespace/dev-apps-jeanphilippels created
PS C:\kubernetes> kubectl get namespace
NAME                STATUS  AGE
default             Active  6m57s
dev-apps-jeanphilippels  Active  9s
kube-node-lease     Active  6m57s
kube-public         Active  6m57s
kube-system         Active  6m58s
local-path-storage  Active  6m49s
PS C:\kubernetes>
```

- Create the Secrets allowing to use private docker repository

```
PS C:\kubernetes> kubectl get secrets
No resources found in default namespace.
PS C:\kubernetes> kubectl get secrets -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl create secret docker-registry devops-dev-sec-jeanphilippels --docker-server=docker.io --docker-username=dockerlsp123 --docker-password=jeanphilippels@hotmail.com --docker-email=jeanphilippels@hotmail.com -n dev-apps-jeanphilippels
secret/devops-dev-sec-jeanphilippels created
PS C:\kubernetes> kubectl get secrets -n dev-apps-jeanphilippels
NAME                TYPE              DATA  AGE
devops-dev-sec-jeanphilippels  kubernetes.io/dockerconfigjson  1      4s
PS C:\kubernetes>
```

- `kubectl create secret docker-registry devops-dev-sec-jeanphilippels --docker-server=docker.io --docker-username=dockerlsp123 --docker-password=<PATAACCESTOKEN> --docker-email=jeanphilippels@hotmail.com -n dev-apps-jeanphilippels`

- Create the service account

```
PS C:\kubernetes> kubectl get sa -n dev-apps-jeanphilippels
NAME      SECRETS  AGE
default   0         6m46s
PS C:\kubernetes> kubectl get sa
NAME      SECRETS  AGE
default   0         13m
PS C:\kubernetes> kubectl apply -f .\serviceaccount.yml
serviceaccount/sa-jpls created
PS C:\kubernetes> kubectl get sa -n dev-apps-jeanphilippels
NAME      SECRETS  AGE
default   0         6m57s
sa-jpls    0         2s
PS C:\kubernetes>
```

- `kubectl apply -f .\serviceaccount.yml -n dev-apps-jeanphilippels`

- Create the role cluster for the service account
- Create the pod's role for the service account
- Bind the role cluster and pod's role to the service account

```
PS C:\kubernetes> kubectl get role -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl get rolebinding -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl apply -f .\role_cluster.yml
clusterrole.rbac.authorization.k8s.io/secret-reader-cluster-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\role_pod.yml
role.rbac.authorization.k8s.io/pod-reader-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\rolebinding-cluster.yml
rolebinding.rbac.authorization.k8s.io/read-secrets-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\rolebinding-sa.yml
rolebinding.rbac.authorization.k8s.io/read-pods-jeanphilippels created
PS C:\kubernetes> kubectl get role -n dev-apps-jeanphilippels
NAME                                CREATED AT
pod-reader-jeanphilippels           2025-03-10T21:36:33Z
PS C:\kubernetes> kubectl get rolebinding -n dev-apps-jeanphilippels
NAME                                ROLE                                AGE
read-pods-jeanphilippels            Role/pod-reader-jeanphilippels      23s
read-secrets-jeanphilippels         Role/secret-reader-jeanphilippels   31s
PS C:\kubernetes> kubectl describe clusterrole secret-reader-cluster-jeanphilippels
Name:                secret-reader-cluster-jeanphilippels
Labels:              <none>
Annotations:         <none>
PolicyRule:
  Resources  Non-Resource URLs  Resource Names  Verbs
  -----
  secrets    []                 []              [get watch list]
PS C:\kubernetes>
```

- `kubectl apply -f .\role_cluster.yml`

- `kubectl apply -f .\role_pod.yml`
- `kubectl apply -f .\rolebinding-cluster.yml`
- `kubectl apply -f .\rolebinding-sa.yml`
- `kubectl get role -n dev-apps-jeanphilippels`
- `kubectl get rolebinding -n dev-apps-jeanphilippels`
- `kubectl describe clusterrole secret-reader-cluster-jeanphilippels`

- **Create the custom devops web application deployment with secret from a docker hub repository**

```
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl apply -f .\deploy_secrets.yml
deployment.apps/devops-deploy-jeanphilippels created
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
```

NAME	READY	STATUS	RESTARTS	AGE
devops-deploy-jeanphilippels-644797c7c-j78jt	0/1	ContainerCreating	0	2s
devops-deploy-jeanphilippels-644797c7c-rfphk	0/1	ContainerCreating	0	2s
devops-deploy-jeanphilippels-644797c7c-wh5tn	0/1	ContainerCreating	0	2s

```
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
```

NAME	READY	STATUS	RESTARTS	AGE
devops-deploy-jeanphilippels-644797c7c-j78jt	1/1	Running	0	14s
devops-deploy-jeanphilippels-644797c7c-rfphk	1/1	Running	0	14s
devops-deploy-jeanphilippels-644797c7c-wh5tn	1/1	Running	0	14s

```
PS C:\kubernetes>
```

```
PS C:\kubernetes> kubectl get deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
devops-deploy-jeanphilippels	3/3	3	3	2m1s

```
PS C:\kubernetes>
```

- `kubectl apply -f .\deploy_secrets.yml`
- `kubectl get pods -n dev-apps-jeanphilippels`
- `kubectl get deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels`

IMPLEMENTING INFRASTRUCTURE AS CODE (IAC)

Terraform

- **Configure terraform local backend: terraform init / terraform plan / terraform apply**
- **Destroy resources provisioned with terraform destroy**
- **Configure terraform backend with Microsoft Azure**

Terraform is an Infrastructure as Code tool that allow to provision automatically infrastructures in the cloud and on-premises. It manages resources and configurations across multiple cloud environments.

- **Configure terraform local backend: terraform init / terraform plan / terraform apply**

The screenshot shows the Visual Studio Code interface with a file explorer on the left and a terminal window at the bottom. The file explorer shows a project structure for 'RESOURCEGROUP' with files like '.terraform', '.terraform.lock.hcl', 'locals.tf', 'main.tf', 'outputs.tf', 'provider.tf', 'variables.tf', and 'versions.tf'. The 'main.tf' file is open in the editor, showing Terraform code for creating an Azure Resource Group and an Azure Storage Account. The terminal window shows the output of the 'terraform init' command, which includes initializing the backend, finding and installing the AzureRM provider, and creating a lock file. The output is as follows:

```
PS C:\terraform\ResourceGroup> terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 4.8.0"...
- Installing hashicorp/azurerm v4.8.0...
- Installed hashicorp/azurerm v4.8.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\terraform\ResourceGroup>
```

The screenshot shows the terminal window with the output of the 'terraform plan' command. The output indicates that Terraform will create an Azure Resource Group and an Azure Storage Account. The output is as follows:

```
PS C:\terraform\ResourceGroup> terraform plan --out tfplanjeanphilippels

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# azurerm_resource_group.mrgpjeanphilippels will be created
+ resource "azurerm_resource_group" "mrgpjeanphilippels" {
  + id       = (known after apply)
  + location = "eastus"
  + name     = "rg-tf-statefiles-jeanphilippels"
}
```

```

# azure_rm_storage_account.mrgpjeanphilippels will be created
+ resource "azure_rm_storage_account" "mrgpjeanphilippels" {
  + access_tier                        = (known after apply)
  + account_kind                      = "StorageV2"
  + account_replication_type          = "LRS"
  + account_tier                      = "Standard"
  + allow_nested_items_to_be_public   = true
  + cross_tenant_replication_enabled   = false
  + default_to_oauth_authentication    = false
  + dns_endpoint_type                 = "Standard"
  + https_traffic_only_enabled         = true
  + id                                = (known after apply)
  + infrastructure_encryption_enabled = false
  + is_hns_enabled                    = false
  + large_file_share_enabled           = (known after apply)
  + local_user_enabled                 = true
  + location                          = "eastus"
  + min_tls_version                    = "TLS1_2"
  + name                              = "tfstatesacntjpls"
  + nfsv3_enabled                     = false
  + primary_access_key                 = (sensitive value)
  + primary_blob_connection_string     = (sensitive value)
  + primary_blob_endpoint              = (known after apply)
  + primary_blob_host                  = (known after apply)
  + primary_blob_internet_endpoint     = (known after apply)
  + primary_blob_internet_host         = (known after apply)
  + primary_blob_microsoft_endpoint    = (known after apply)
  + primary_blob_microsoft_host        = (known after apply)
  + primary_connection_string          = (sensitive value)
  + primary_dfs_endpoint               = (known after apply)
  + primary_dfs_host                   = (known after apply)
  + primary_dfs_internet_endpoint      = (known after apply)
  + primary_dfs_internet_host          = (known after apply)
  + primary_dfs_microsoft_endpoint     = (known after apply)
  + primary_dfs_microsoft_host         = (known after apply)
  + primary_file_endpoint              = (known after apply)
  + primary_file_host                  = (known after apply)
  + primary_file_internet_endpoint     = (known after apply)
  + primary_file_internet_host         = (known after apply)
  + primary_file_microsoft_endpoint    = (known after apply)
  + primary_file_microsoft_host        = (known after apply)
  + primary_location                   = (known after apply)
  + primary_queue_endpoint             = (known after apply)
  + primary_queue_host                 = (known after apply)
  + primary_queue_microsoft_endpoint   = (known after apply)
  + primary_queue_microsoft_host       = (known after apply)
  + primary_table_endpoint             = (known after apply)
  + primary_table_host                 = (known after apply)
  + primary_table_microsoft_endpoint   = (known after apply)

```

```

+ secondary_file_host = (known after apply)
+ secondary_file_internet_endpoint = (known after apply)
+ secondary_file_internet_host = (known after apply)
+ secondary_file_microsoft_endpoint = (known after apply)
+ secondary_file_microsoft_host = (known after apply)
+ secondary_location = (known after apply)
+ secondary_queue_endpoint = (known after apply)
+ secondary_queue_host = (known after apply)
+ secondary_queue_microsoft_endpoint = (known after apply)
+ secondary_queue_microsoft_host = (known after apply)
+ secondary_table_endpoint = (known after apply)
+ secondary_table_host = (known after apply)
+ secondary_table_microsoft_endpoint = (known after apply)
+ secondary_table_microsoft_host = (known after apply)
+ secondary_web_endpoint = (known after apply)
+ secondary_web_host = (known after apply)
+ secondary_web_internet_endpoint = (known after apply)
+ secondary_web_internet_host = (known after apply)
+ secondary_web_microsoft_endpoint = (known after apply)
+ secondary_web_microsoft_host = (known after apply)
+ sftp_enabled = false
+ shared_access_key_enabled = true
+ table_encryption_key_type = "Service"
+ tags = {
  + "environment" = "dev"
  + "owner" = "IT"
}

+ blob_properties (known after apply)

+ network_rules (known after apply)

+ queue_properties (known after apply)

+ routing (known after apply)

+ share_properties (known after apply)
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:

+ storage_account_name = "tfstatesacctjpls"

Saved the plan to: tfplanjeanphilippls

To perform exactly these actions, run the following command to apply:
 terraform apply "tfplanjeanphilippls"

```
PS C:\terraform\ResourceGroup> terraform apply -tfplanjeanphilippls
azurerm_resource_group.mrgjeanphilippls: Creating...
azurerm_resource_group.mrgjeanphilippls: Still creating... [10s elapsed]
azurerm_resource_group.mrgjeanphilippls: Creation complete after 13s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilippls]
azurerm_storage_account.mrgjeanphilippls: Creating...
azurerm_storage_account.mrgjeanphilippls: Still creating... [10s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [20s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [30s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [40s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [50s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [60s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [70s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [80s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [90s elapsed]
azurerm_storage_account.mrgjeanphilippls: Still creating... [100s elapsed]
azurerm_storage_account.mrgjeanphilippls: Creation complete after 1m17s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilippls/providers/Microsoft.Storage/storageAccounts/tfstatesacctjpls]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
storage_account_name = "tfstatesacctjpls"
PS C:\terraform\ResourceGroup> ]
```

```

1 {
2   "version": 4,
3   "terraform_version": "1.11.0",
4   "serial": 3,
5   "lineage": "86461dbf-9eb7-31a6-8c00-fc3808755bcd",
6   "outputs": {
7     "storage_account_name": {
8       "value": "tfstatejeanphilipps",
9       "type": "string"
10    }
11  },
12  "resources": [
13    {
14      "mode": "managed",
15      "type": "azurem_resource_group",
16      "name": "rg-jeanphilipps",
17      "provider": ["registry.terraform.io/hashicorp/azurerm"],
18      "instances": [
19        {
20          "schema_version": 0,
21          "attributes": {
22            "id": "/subscriptions/8de90c41-7afc-4788-9660-7d97d0c82cab/resourcegroups/rg-tf-statefiles-jeanphilipps",
23            "location": "eastus",
24            "managed_by": "",
25            "name": "rg-tf-statefiles-jeanphilipps",
26            "tags": null,
27            "timeouts": null
28          },
29          "sensitive_attributes": [],
30          "private": "eyJjbWZwczpkIjVzPHJlTEx2YUQYAC0BNPQH2j3N2NPYzA1oms1Y3V3YXItIjo1bDAwPDdwPDwPdWwCj3Z0ciIGU0SjBWPdAwPDdwPDwPdAsIn1VQ10jPHwPdAwPDwPdwcUci0kXYXR1Ij0"
31        }
32      ]
33    },
34    {
35      "mode": "managed",
36      "type": "azurem_storage_account",
37      "name": "mrpjjeanphilipps",
38      "provider": ["registry.terraform.io/hashicorp/azurerm"],
39      "instances": [
40        {
41          "schema_version": 4,
42          "attributes": {

```


The top screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation indicates the path: Home > Resource groups > rg-tf-statefiles-jeanphilipps. The page title is 'Resource groups' with a search icon and a close button. Below the title, there are options to 'Create', 'Manage view', 'Refresh', 'Export to CSV', 'Open query', and 'Assign tags'. A search bar contains 'rg-tf-statefiles'. Below the search bar, there are filters for 'Subscription equals all' and 'Location equals all'. The results show one resource group: 'rg-tf-statefiles-jeanphilipps' located in 'East US' under the 'Azure for Students' subscription.

The bottom screenshot shows the details of the storage account 'tfstatesacctjpls'. The breadcrumb navigation is: Home > Resource groups > rg-tf-statefiles-jeanphilipps > tfstatesacctjpls. The page title is 'tfstatesacctjpls' with a search icon and a close button. Below the title, there are options to 'Upload', 'Open in Explorer', 'Delete', 'Move', 'Refresh', 'Open in mobile', 'CLI / PS', and 'Feedback'. The 'Overview' tab is selected. It shows the resource group 'rg-tf-statefiles-jeanphilipps' and the location 'eastus'. The subscription is 'Azure for Students'. The subscription ID is '8de89c41-7afc-4788-9666-7d97d6c02cab'. The disk state is 'Available'. The tags are 'environment: dev' and 'owner: IT'. The 'Performance' section shows 'Standard' performance, 'Locally-redundant storage (LRS)' replication, 'StorageV2 (general purpose v2)' account kind, and 'Succeeded' provisioning state. The 'Properties' section shows 'Blob service' settings: Hierarchical namespace (Disabled), Default access tier (Hot), Blob anonymous access (Enabled), Blob soft delete (Disabled), Container soft delete (Disabled), Versioning (Disabled), Change feed (Disabled), NFS v3 (Disabled), and Allow cross-tenant replication (Disabled). The 'Security' section shows 'Require secure transfer for REST API operations' (Enabled), 'Storage account key access' (Enabled), 'Minimum TLS version' (Version 1.2), and 'Infrastructure encryption' (Disabled). The 'Networking' section shows 'Allow access from' (All networks), 'Private endpoint connections' (0), and 'Network routing' (Microsoft network routing).

- **Destroy resources provisioned with “terraform destroy”**

```

C:\terraform>terraform plan -destroy -target=tfstatesacctjpls-jeanphilipps
azure_resource_group.azrgjeanphilipps: Refreshing state... [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps]
azure_storage_account.azsgjeanphilipps: Refreshing state... [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps/providers/Microsoft.Storage/storageAccounts/tfstatesacctjpls]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# azure_resource_group.azrgjeanphilipps will be destroyed
resource "azure_resource_group" "azrgjeanphilipps" {
  id      = "/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps" -> null
  location = "eastus" -> null
  name     = "rg-tf-statefiles-jeanphilipps" -> null
  tags    = {} -> null
  # (4 unchanged attribute hidden)
}

# azure_storage_account.azsgjeanphilipps will be destroyed
resource "azure_storage_account" "azsgjeanphilipps" {
  access_tier                = "net" -> null
  account_kind               = "Storage" -> null
  account_replication_type   = "LRS" -> null
  account_tier               = "Standard" -> null
  allow_nested_items_to_be_public = true -> null
  cross_tenant_replication_enabled = false -> null
  default_to_outh_authentication = false -> null
  dns_endpoint_type          = "Standard" -> null
  https_traffic_only_enabled = true -> null
  id                         = "/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps/providers/Microsoft.Storage/storageAccounts/tfstatesacctjpls" -> null
  infrastructure_encryption_enabled = false -> null
  is_hns_enabled             = false -> null
  large_file_share_enabled   = true -> null
  local_user_enabled         = true -> null
  location                   = "eastus" -> null
  min_tls_version            = "TLS1_2" -> null
  name                       = "tfstatesacctjpls" -> null
  nfv1_enabled               = false -> null
  primary_access_key         = (sensitive value) -> null
  primary_blob_connection_string = (sensitive value) -> null
  primary_blob_endpoint       = "https://tfstatesacctjpls.blob.core.windows.net/" -> null
  primary_blob_host           = "tfstatesacctjpls.blob.core.windows.net" -> null
  primary_connection_string   = (sensitive value) -> null
  primary_dfs_endpoint        = "https://tfstatesacctjpls.dfs.core.windows.net/" -> null
  primary_dfs_host            = "tfstatesacctjpls.dfs.core.windows.net" -> null
  primary_file_share_endpoint = "https://tfstatesacctjpls.file.core.windows.net/" -> null
}

```

```

- blob_properties {
  - change_feed_enabled      = false -> null
  - change_feed_retention_in_days = 0 -> null
  - last_access_time_enabled   = false -> null
  - versioning_enabled        = false -> null
  # (1 unchanged attribute hidden)
}

- queue_properties {
  - hour_metrics {
    - enabled      = false -> null
    - include_apis = false -> null
    - retention_policy_days = 0 -> null
    - version      = "1.0" -> null
  }
  - logging {
    - delete      = false -> null
    - read        = false -> null
    - retention_policy_days = 0 -> null
    - version      = "1.0" -> null
    - write       = false -> null
  }
  - minute_metrics {
    - enabled      = false -> null
    - include_apis = false -> null
    - retention_policy_days = 0 -> null
    - version      = "1.0" -> null
  }
}

- share_properties {
  - retention_policy {
    - days = 7 -> null
  }
}
}

```

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:

- storage_account_name = "tfstatesacctjpls" -> null

Saved the plan to: tfplandestroy_jeanphilippels

To perform exactly these actions, run the following command to apply:

```
terraform apply "tfplandestroy_jeanphilippels"
```

PS C:\terraform\ResourceGroup>

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups (jeanphilippels@gmail.com@microsoft.com)

+ Create Manage view Refresh Export to CSV Open query Assign tags

rg-tf-statefiles Subscription equals all Location equals all Add filter

Showing 0 to 0 of 0 records. No grouping List view

Name Subscription Location

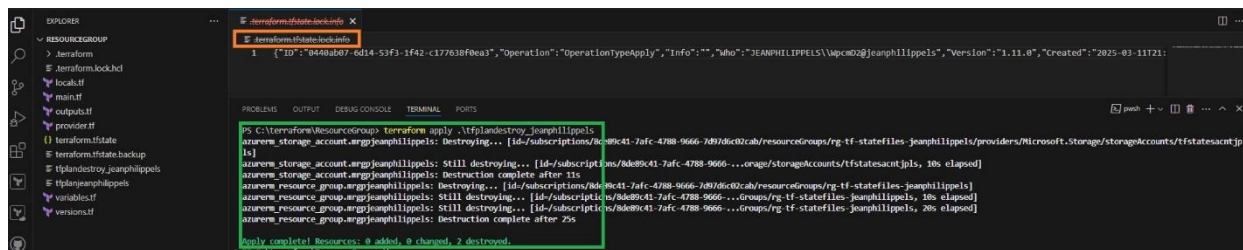
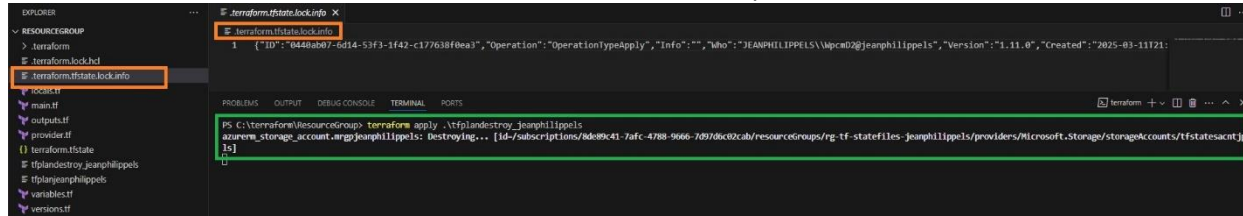
No resource groups match your filters

Resource groups provide a logical container to manage and organize Azure resources, simplifying administration and enabling efficient resource management.

+ Create Clear filters

Learn more

When a terraform apply is executing a `.terraform.tfstate.lock.info` is created in order to lock the tfstate so no one can execute at the same time a terraform command and corrupt the infrastructure.

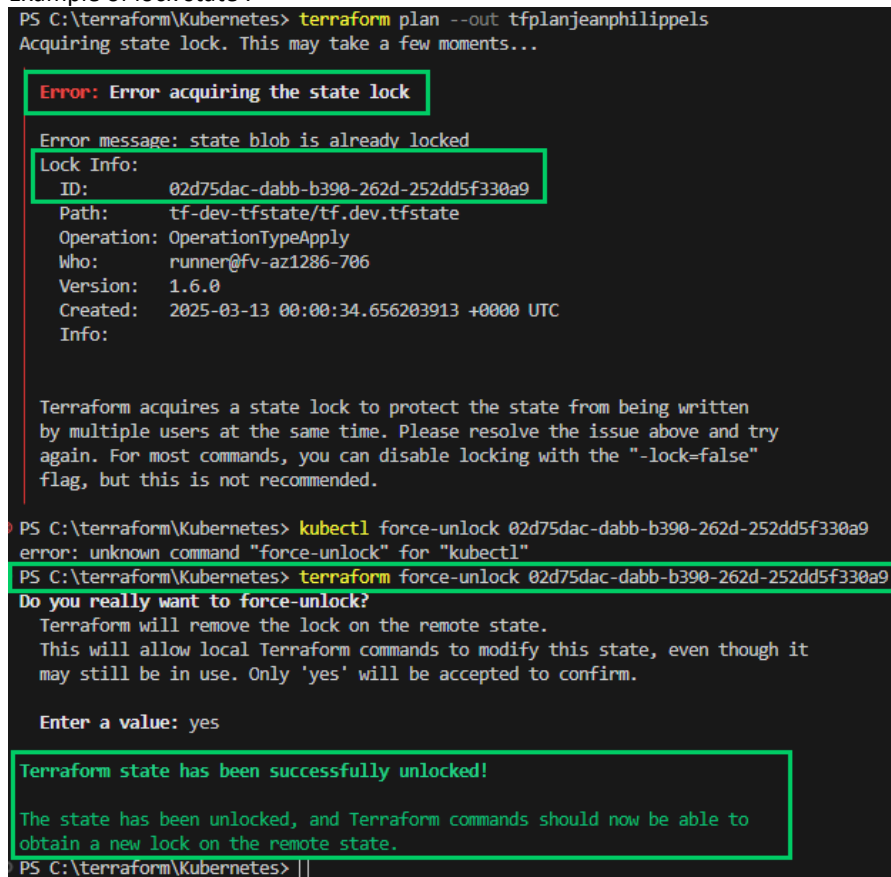


- **Configure terraform backend with Microsoft Azure**

State file can contain secrets, keeping it in Azure storage enhance security.

This will ensure that the latest version of the state file is used, while locking other user to deploy.

Example of lock state :



We provision the backend with terraform on Microsoft Azure.

- Provision storage account and container without backend block = `terraform.tfstate` locally

- After provisioning: add the backend block with references to the resources just created
- Run terraform init.
- Terraform will ask to migrate the local state file into Microsoft Azure Storage

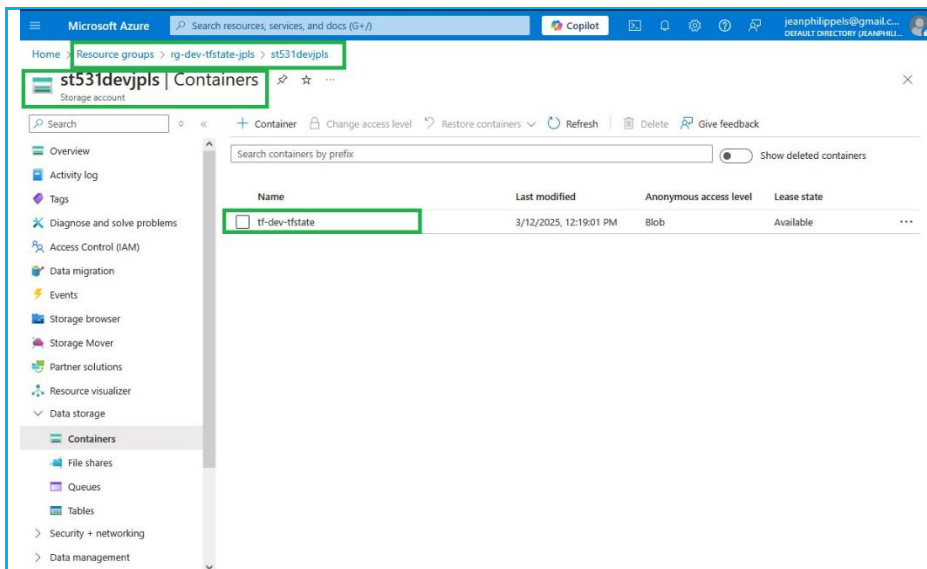
```
PS C:\terraform\ResourceGroup> terraform plan -out tfplanjeanphilipps

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

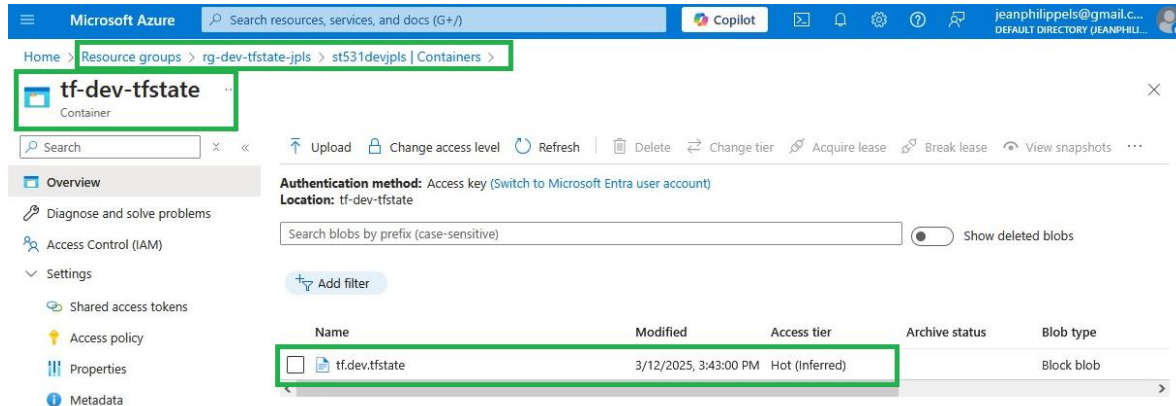
Terraform will perform the following actions:

# azurem_resource_group.mrgpjeanphilipps will be created
+ resource "azurem_resource_group" "mrgpjeanphilipps" {
  + id           = (known after apply)
  + location     = "eastus"
  + name        = "rg-dev-tfstate-jpls"
}

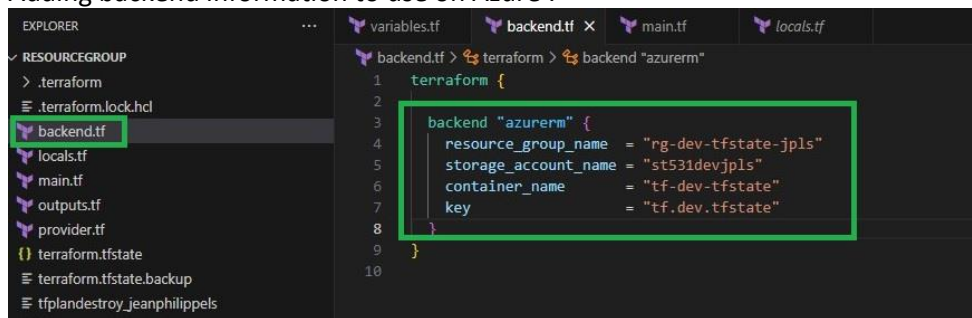
# azurem_storage_account.mrgpjeanphilipps will be created
+ resource "azurem_storage_account" "mrgpjeanphilipps" {
  + access_tier                = (known after apply)
  + account_kind               = "StorageV2"
  + account_replication_type   = "LRS"
  + account_tier                = "Standard"
  + allow_nested_items_to_be_public = true
  + cross_tenant_replication_enabled = false
  + default_to_oauth_authentication = false
  + dns_endpoint_type          = "Standard"
  + https_traffic_only_enabled   = true
  + id                         = (known after apply)
  + infrastructure_encryption_enabled = false
  + is_hns_enabled              = false
  + large_file_share_enabled     = (known after apply)
  + local_user_enabled          = true
  + location                   = "eastus"
  + min_tls_version             = "TLS1_2"
  + name                       = "st53idevjpls"
  + nfsv3_enabled               = false
  + primary_access_key           = (sensitive value)
  + primary_blob_connection_string = (sensitive value)
  + primary_blob_endpoint        = (known after apply)
  + primary_blob_host             = (known after apply)
  + primary_blob_internet_endpoint = (known after apply)
  + primary_blob_internet_host    = (known after apply)
  + primary_blob_microsoft_endpoint = (known after apply)
  + primary_blob_microsoft_host   = (known after apply)
  + primary_connection_string     = (sensitive value)
  + primary_dfs_endpoint          = (known after apply)
  + primary_dfs_host              = (known after apply)
  + primary_dfs_internet_endpoint = (known after apply)
  + primary_dfs_internet_host     = (known after apply)
  + primary_dfs_microsoft_endpoint = (known after apply)
  + primary_dfs_microsoft_host    = (known after apply)
  + primary_file_endpoint         = (known after apply)
  + primary_file_host             = (known after apply)
  + primary_file_internet_endpoint = (known after apply)
  + primary_file_internet_host    = (known after apply)
}
```

Tf State file is stored in the Azure container with the "tf.dev.tfstate" name.



Adding backend information to use on Azure :



The screenshot shows the Visual Studio Code interface with the Explorer pane on the left displaying the project structure. The main editor shows the `backend.tf` file with the following content:

```

1 terraform {
2
3   backend "azurerm" {
4     resource_group_name = "rg-dev-tfstate-jpls"
5     storage_account_name = "st531devjpls"
6     container_name      = "tf-dev-tfstate"
7     key                 = "tf.dev.tfstate"
8   }
9 }
10

```

The TERMINAL pane at the bottom shows the output of the `terraform init` command:

```

PS C:\terraform\ResourceGroup> terraform init
Initializing the backend...
Acquiring state lock. This may take a few moments...
Do you want to copy existing state to the new backend?
Pre-existing state was found while migrating the previous "local" backend to the
newly configured "azurerm" backend. No existing state was found in the newly
configured "azurerm" backend. Do you want to copy this state to the new "azurerm"
backend? Enter "yes" to copy and "no" to start with an empty state.
Enter a value: yes
Releasing state lock. This may take a few moments...
Successfully configured the backend "azurerm"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing provider plugins...
- Reusing previous version of hashicorp/azurerm from the dependency lock file
- Using previously-installed hashicorp/azurerm v4.8.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\terraform\ResourceGroup>

```

To verify the setup, the local instance of **terraform.tfstate** file has been deleted .
We run command `terraform state list`. The output is as follow:

The screenshot shows the Visual Studio Code interface with the Explorer pane on the left displaying the project structure. The main editor shows the `backend.tf` file with the same content as the previous screenshot.

The TERMINAL pane at the bottom shows the output of the `terraform state list` command:

```

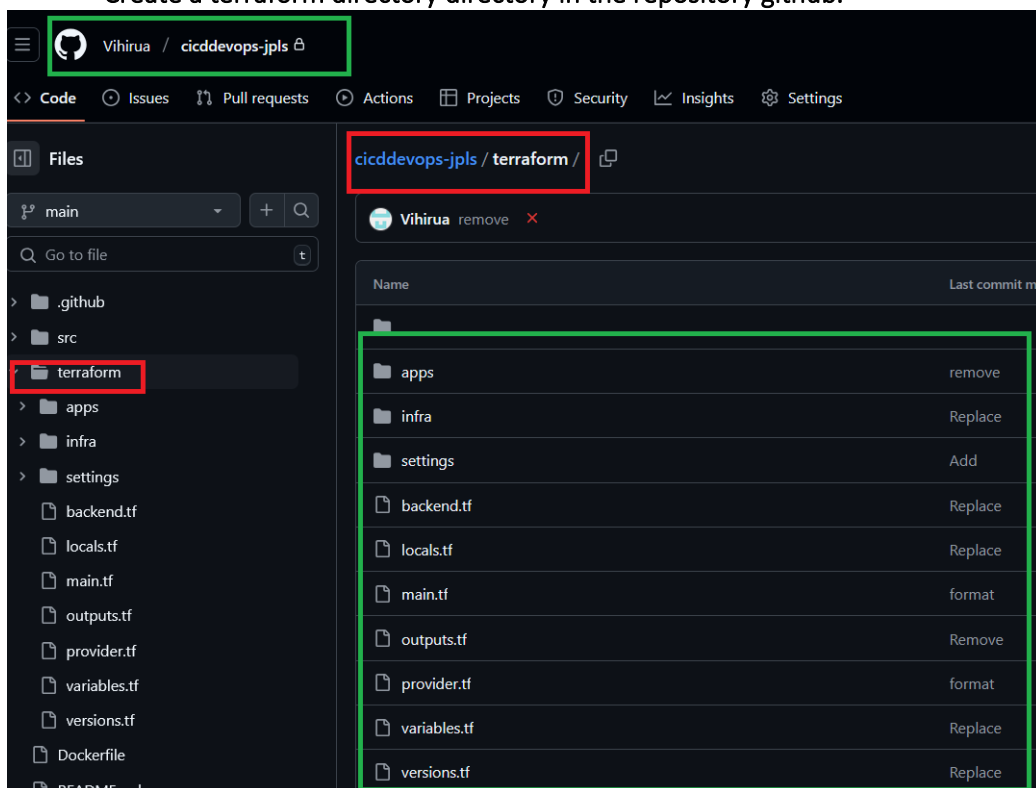
PS C:\terraform\ResourceGroup> terraform state list
azurerm_resource_group.mrgpjeanphilippels
azurerm_storage_account.mrgpjeanphilippels
azurerm_storage_container.terraformbackend
PS C:\terraform\ResourceGroup>

```

Deployment in Azure Kubernetes Cluster through Terraform

- **Deploy the container using Kubernetes / Apply Kubernetes Configuration**
 - Create a terraform directory in the repository github.
 - *Configure Azure Kubernetes Services*
- *Provision the Azure Kubernetes Services*
- *Create the custom namespace for devops applications*
- *Create the Secrets allowing to use private docker repository*
- *Create the Service account*
- *Create the role cluster for the service account*
- *Bind the role cluster to the service account*
- *Create the custom devops web application deployment with secret from a docker hub repository*
- *Create the service custom devops web application deployment*
- *Provision the Apps in Azure Kubernetes Services*
- *Testing the devops web deployment*

- *Create a terraform directory in the repository github.*



- *Configure Azure Kubernetes Services*

In order terraform to access Microsoft Azure, a Service Principal Name has to be created. Contributor Role is giving to the Service Principal Name.



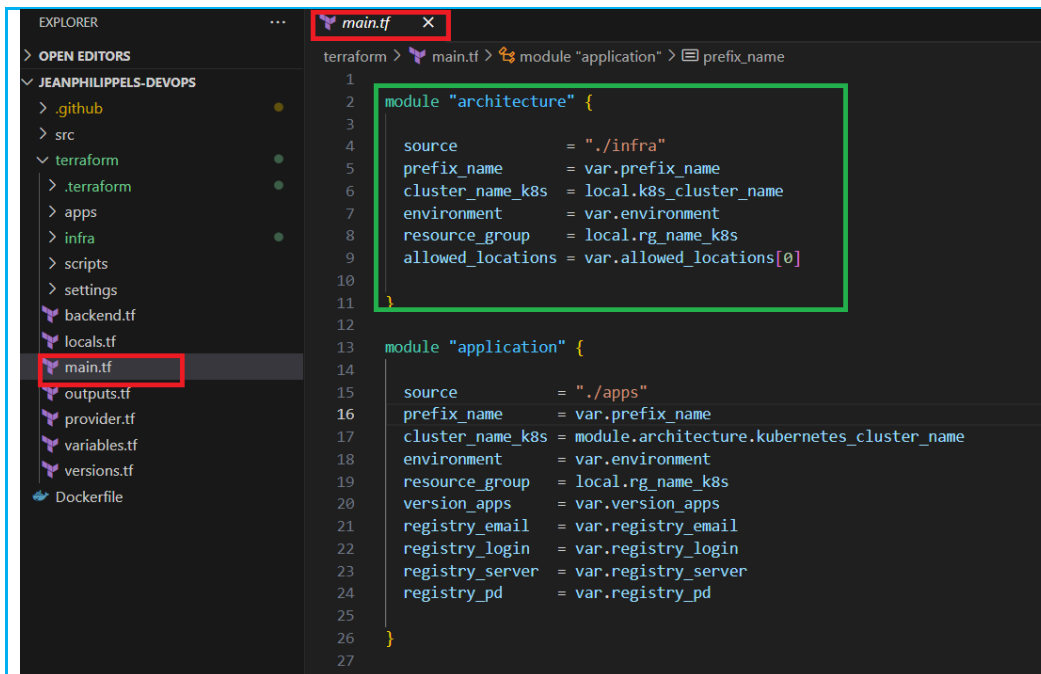
The screenshot shows a VS Code editor with a Terraform configuration file named `provider.tf`. The file contains two provider blocks. The `azurerm` block is highlighted with a green box. The `kubernetes` block is also visible below it. The left sidebar shows the file explorer with `provider.tf` selected.

```

1
2 provider "azurerm" {
3   subscription_id = "8de89c41-7afc-4788-9666-7d97d6c02cab"
4   tenant_id      = "7ba26a6d-8f43-4467-b442-67df22a1f7d"
5   features {
6
7     key_vault {
8       purge_soft_delete_on_destroy = false
9     }
10    resource_group {
11      prevent_deletion_if_contains_resources = false
12    }
13  }
14 }
15
16 provider "kubernetes" {
17
18   host = azurerm_kubernetes_cluster.default.kube_admin_config.0.host
19   client_certificate = base64decode(azurerm_kubernetes_cluster.default.kube_admin_config.0.client_certificate)
20   client_key         = base64decode(azurerm_kubernetes_cluster.default.kube_admin_config.0.client_key)
21   cluster_ca_certificate = base64decode(azurerm_kubernetes_cluster.default.kube_admin_config.0.cluster_ca_certificate)
22 }

```

Project : DevOps implementation Document | version 1.0 | Final Release



The screenshot shows the VS Code interface with the Explorer sidebar on the left and the main editor on the right. In the Explorer, the file `main.tf` is selected and highlighted with a red box. The breadcrumb at the top of the editor reads: `terraform > main.tf > module "application" > prefix_name`. The main editor displays the content of `main.tf`. A green box highlights the `module "architecture"` block, which is defined as follows:

```
1 module "architecture" {
2
3   source           = "../infra"
4   prefix_name      = var.prefix_name
5   cluster_name_k8s = local.k8s_cluster_name
6   environment      = var.environment
7   resource_group   = local.rg_name_k8s
8   allowed_locations = var.allowed_locations[0]
9 }
10
11
12 module "application" {
13
14   source           = "../apps"
15   prefix_name      = var.prefix_name
16   cluster_name_k8s = module.architecture.kubernetes_cluster_name
17   environment      = var.environment
18   resource_group   = local.rg_name_k8s
19   version_apps     = var.version_apps
20   registry_email   = var.registry_email
21   registry_login   = var.registry_login
22   registry_server  = var.registry_server
23   registry_pd      = var.registry_pd
24 }
25
26
27
```

- **Module architecture:**

The variables of the module are the parameters that need to be fill up when calling in the main root directory of terraform :

```
PS C:\Users\Wpcmd2\Documents\Github\jeanphilippe\repos\terraform> terraform apply --target=module.architecture --var=filesettings/dev/variables.tfvars --auto-approve
```

```
Acquiring state lock. This may take a few moments...
module.architecture.data.azure_service_principal.splpaact: Reading...
module.architecture.data.azuread_client_config.current: Reading...
module.architecture.data.azure_service_principal.splpacr: Reading...
module.architecture.data.azure_group.groupdevops: Reading...
module.architecture.data.azuread_client_config.current: Read complete after 0s [id=/ba7b6d6d-bf43-4467-b447-d7fd221af7d-49642172-7e15-4446-a15e-bdf3ef8ca293-aeb41d91-48fc-4f8c-b320-33346dd37471]
module.architecture.data.azure_service_principal.splpacr: Read complete after 1s [id=/servicePrincipals/aeb41d91-48fc-4f8c-b320-33346dd37471]
module.architecture.data.azure_service_principal.splpacr: Read complete after 1s [id=/servicePrincipals/b6ab327a-b37c-4f32-879f-f614d982745]
module.architecture.data.azure_group.groupdevops: Read complete after 1s [id=/groups/6658a60b-0395-46d3-b6c2-bbebf1afe0e]
module.architecture.data.azuread_client_config.current: Reading...
module.architecture.data.azuread_client_config.current: Reading...
module.architecture.data.azure_role_definition.builtin: Reading...
module.architecture.data.azure_subscription.primary: Reading...
module.architecture.data.azuread_client_config.current: Read complete after 0s [id=YxpZxS0Q0zNuZmncYnjbglB1LnRZD0B0TYQMjE3MlO3TE1LTQNDAVTYE1ZS1ZDNmZTg0V2YTEOTH7b2JqZJmWSHQYVMYNDFKOTEndhmY06ZjhUWtHjAhtMhNZkZDM3NDc03N1VmjcmJwLWVklbPtkHzTSyV-qOLtdbmZNDC4OC8NSYv2LTkdOtGdnNmWmhny1h8Z5NBhNRZD83YmEZC2B4ZjQLTG0MTCjY3QMlBZ82R2ZjYTFmMQZ]
module.architecture.data.azuread_client_config.current: Read complete after 0s [id=YxpZxS0Q0zNuZmncYnjbglB1LnRZD0B0TYQMjE3MlO3TE1LTQNDAVTYE1ZS1ZDNmZTg0V2YTEOTH7b2JqZJmWSHQYVMYNDFKOTEndhmY06ZjhUWtHjAhtMhNZkZDM3NDc03N1VmjcmJwLWVklbPtkHzTSyV-qOLtdbmZNDC4OC8NSYv2LTkdOtGdnNmWmhny1h8Z5NBhNRZD83YmEZC2B4ZjQLTG0MTCjY3QMlBZ82R2ZjYTFmMQZ]
module.architecture.data.azuread_client_config.current: Read complete after 0s [id=YxpZxS0Q0zNuZmncYnjbglB1LnRZD0B0TYQMjE3MlO3TE1LTQNDAVTYE1ZS1ZDNmZTg0V2YTEOTH7b2JqZJmWSHQYVMYNDFKOTEndhmY06ZjhUWtHjAhtMhNZkZDM3NDc03N1VmjcmJwLWVklbPtkHzTSyV-qOLtdbmZNDC4OC8NSYv2LTkdOtGdnNmWmhny1h8Z5NBhNRZD83YmEZC2B4ZjQLTG0MTCjY3QMlBZ82R2ZjYTFmMQZ]
module.architecture.data.azure_subscription.primary: Read complete after 0s [id=/subscriptions/4081c7-74fc-4868-966b-7d97de6bbac]
module.architecture.data.azure_role_definition.akademinorole: Read complete after 0s [id=/providers/Microsoft.Authorization/roleDefinitions/scdcd727-384e-4bdc-b6d4-f660fab81ea7]
module.architecture.data.azure_role_definition.akademinorole: Read complete after 1s [id=/providers/Microsoft.Authorization/roleDefinitions/b1ff9b4d-80eb-8693973ca19b]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

```
+ create
<- read (data_resources)
```

Terraform will perform the following actions:

```
# module.architecture.data.azuread_group.aksigroup will be read during apply
# (depends on a resource or a module with changes pending)
<= data "azuread_group" "aksigroup" {
  + assignable_to_role           = (known after apply)
  + auto_subscribe_new_members  = (known after apply)
  + behaviors                    = (known after apply)
  + description                  = (known after apply)
  + display_name                 = "k8s_admin"
  + dynamic_membership           = (known after apply)
  + external_senders_allowed     = (known after apply)
  + hide_from_address_lists     = (known after apply)
```

```
+ content_sha512 = (known after apply)
+ content_sha1 = (known after apply)
+ content_sha256 = (known after apply)
+ content_sha512 = (known after apply)
+ directory_permission = "0777"
+ file_permission = "0777"
+ filename = "/kubernetes"
+ id = (known after apply)
}

# module.architecture.random_pet.ssh_key_name will be created
+ resource "random_pet" "ssh_key_name" {
+   id = (known after apply)
+   length = 2
+   prefix = "ssh"
+   # (1 unchanged attribute hidden)
}
```

Plan: 22 to add, 0 to change, 0 to destroy.

changes to Outputs:

```
+ aks_msi_principal_id = (known after apply)
+ client_certificate    = (sensitive value)
+ client_key            = (sensitive value)
+ cluster_ca_certificate = (sensitive value)
+ cluster_name          = "k8s-dev-cluster"
+ cluster_username      = (sensitive value)
+ host                  = (sensitive value)
+ kube_config            = (sensitive value)
```

```

One config [Sensitive Value]
module.architecture.random_pet_ssh_key_name: Creating...
module.architecture.random_pet_ssh_key_name: Creation complete after 0s [id=ssh-humblestrimp]
module.architecture.azure_rm_role_assignment.aks_sp_role_assignment: Creating...
module.architecture.azure_rm_resource_group.rgaks: Creating...
module.architecture.azuread_group.aks_user: Creating...
module.architecture.azuread_group.aks_admin: Creating...
module.architecture.azure_rm_resource_group.rgaks: Still creating... [18s elapsed]
module.architecture.azure_rm_resource_group.rgaks: Creation complete after 15s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97dc62cab/resourceGroups/rg-dev-k8s-jpls]
module.architecture.azurerm_subscription.snbuplg_key_creating: Creating...

```

```

module.architecture.azure_rm_kubernetes_cluster.default: Still creating... [5m30s elapsed]
module.architecture.azure_rm_kubernetes_cluster.default: Creation complete after 5m36s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-k8s-jpls/providers/Microsoft.ContainerService/managedClusters/k8s-dev-cluster]
module.architecture.azuread_group_member.groupmemberspaks_msi: Creating...
module.architecture.kubernetes_cluster_role_binding.act_bind: Creating...
module.architecture.local_file.kubeconfig: Creating...
module.architecture.kubernetes_cluster_role.actallow: Creating...
module.architecture.local_file.kubeconfig: Creation complete after 0s [id=4ed9bf0358a7c541e55958f1176c8306ce1307c4]
module.architecture.azuread_group_member.groupmemberspaks_msi: Creation complete after 1s [id=436fa8a1-831a-4ebe-8618-7afa59ab06e3/member/a210dbbd-2f7e-47c1-83fb-8db8359adbca]
module.architecture.kubernetes_cluster_role.actallow: Creation complete after 1s [id=managed-cluster]
module.architecture.kubernetes_cluster_role_binding.act_bind: Creation complete after 1s [id=managed-cluster-bind]

Warning: Resource targeting is in effect

You are creating a plan with the -target option, which means that the result of this plan may not represent all of the changes requested by the current configuration.

The -target option is not for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Warning: Applied changes may be incomplete

The plan was created with the -target option in effect, so some changes requested in the configuration may have been ignored and the output values may not be fully updated. Run the following command to verify that no other changes are pending:

    terraform plan

Note that the -target option is not suitable for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Releasing state lock. This may take a few moments...

Apply complete! Resources: 22 added, 0 changed, 0 destroyed.

Outputs:

aks_msi_principal_id = "a210dbbd-2f7e-47c1-83fb-8db8359adbca"
client_certificate = <sensitive>
client_key = <sensitive>
cluster_ca_certificate = <sensitive>
cluster_name = "k8s-dev-cluster"
cluster_username = <sensitive>
host = <sensitive>
kube_config = <sensitive>

```

k8s-dev-cluster Kubernetes service

Search

Overview

Activity log

Access control (IAM)

Tags

Monitor

Diagnose and solve problems

Microsoft Defender for Cloud (preview)

Cost analysis

Resource visualizer

Kubernetes resources

- Namespaces
- Workloads
- Services and ingress
- Storage
- Configuration
- Custom resources

Essentials

Resource group	: rg-dev-k8s-jpls	Kubernetes version	: 1.32.0
Power state	: Running	API server address	: k8sdevdns-kwmjxu7yhcp.eastus.azmk8s.io
Cluster operation status	: Succeeded	Network configuration	: Azure CNI Overlay
Subscription	: Azure for Students	Node pools	: 1 node pool
Location	: East US	Container registries	: Attach a registry

Subscription ID : 8de89c41-7afc-4788-9666-7d97d6c02cab

Tags (edit) : company: OldSchool cost_center: cc-123 environment: dev managed_by: terraform

Get started **Properties** Monitoring Recommendations

Kubernetes services

Encryption type	Encryption at-rest with a platform-managed key
Virtual node pools	Not enabled

Node pools

Node pools	1 node pool
Kubernetes versions	1.32.0
Node sizes	Standard_D2_v2

Networking

API server address	k8sdevdns-kwmjxu7yhcp.eastus.azmk8s.io
Network configuration	Azure CNI Overlay
Pod CIDR	10.244.0.0/16
Service CIDR	10.0.0.0/16
DNS service IP	10.0.0.10
Cilium dataplane	Not enabled
Network Policy	None

Azure Kubernetes Services relies on Virtual Machine Scale Set which a group of load balanced Virtual Machines.

aks-k8sdevnode-32010145-vmss
Virtual machine scale set

Overview

Essentials

Resource group (move) : [rg-dev-k8s-jpls-nrg](#)
 Status : 2 out of 2 succeeded
 Location : East US
 Subscription (move) : [Azure for Students](#)
 Subscription ID : 8de89c41-7afc-4788-9666-7d97d6c02cab

Operating system : Linux
 Size : Standard_D2_v2 (2 instances)
 Public IP address : [172.171.57.60](#)
 Public address (IPv6) : -
 Virtual network/subnet : [aks-vnet-12231162/aks-subnet](#)
 Orchestration mode : Uniform
 Time created : 3/14/2025, 10:24 PM UTC

Tags (edit) : aks-managed-azure-cni-overlay : true aks-managed-consolidated-additio... : e77fda75-0122-11f0-9f7b-c...

Properties Monitoring Capabilities (6) Recommendations Tutorials

Virtual machine profile
 Operating system : Linux
 Capacity reservation group : -
 Hibernation : Disabled

Azure Spot
 Azure Spot : Disabled

Availability + scaling
 Availability zone : -
 Proximity placement group : -
 Colocation status : -

Status
 Provisioning state : Succeeded
 Power state : 2 out of 2 running

Networking
 Public IP address : -
 Public IP address (IPv6) : -
 Virtual network/subnet : [aks-vnet-12231162/aks-subnet](#)

Size
 Size : Standard_D2_v2
 vCPUs : 2

rg-dev-k8s-jpls-nrg
Resource group

Resources

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 7 of 7 records. Show hidden types No grouping List view

Name	Type	Location
416ac108-8245-403b-bbb7-ce34abc7dc80	Public IP address	East US
aks-agentpool-12231162-nsg	Network security group	East US
aks-k8sdevnode-34073874-vmss	Virtual machine scale set	East US
aks-vnet-12231162	Virtual network	East US
azurepolicy-k8s-dev-cluster	Managed Identity	East US
k8s-dev-cluster-agentpool	Managed Identity	East US
kubernetes	Load balancer	East US

- **Create the custom namespace for devops applications**

The screenshot displays a development environment with two main components: a code editor and the Microsoft Azure portal.

Code Editor (Top): The file explorer on the left shows a project structure for 'JEANPHILIPPELS-DEVOPS'. The file 'aks-apps.tf' is selected and highlighted with a red box. The code editor shows the Terraform configuration for a Kubernetes namespace, with the resource definition highlighted by a green box:

```

1 # ----- AKS APPLICATIONS ----- #
2
3 #Kubernetes namespace to hold application
4 resource "kubernetes_namespace" "tf_k8s_ns" {
5
6   metadata {
7     name = local.k8s_namespace_devops_apps
8     labels = {
9       environment = local.environment_tags["environment"]
10      managed_by   = local.default_tags["managed_by"]
11      type         = local.default_tags["type"]
12      department   = local.default_tags["department"]
13    }
14  }
15 }
16
17 # ----- Secrets ----- #
18

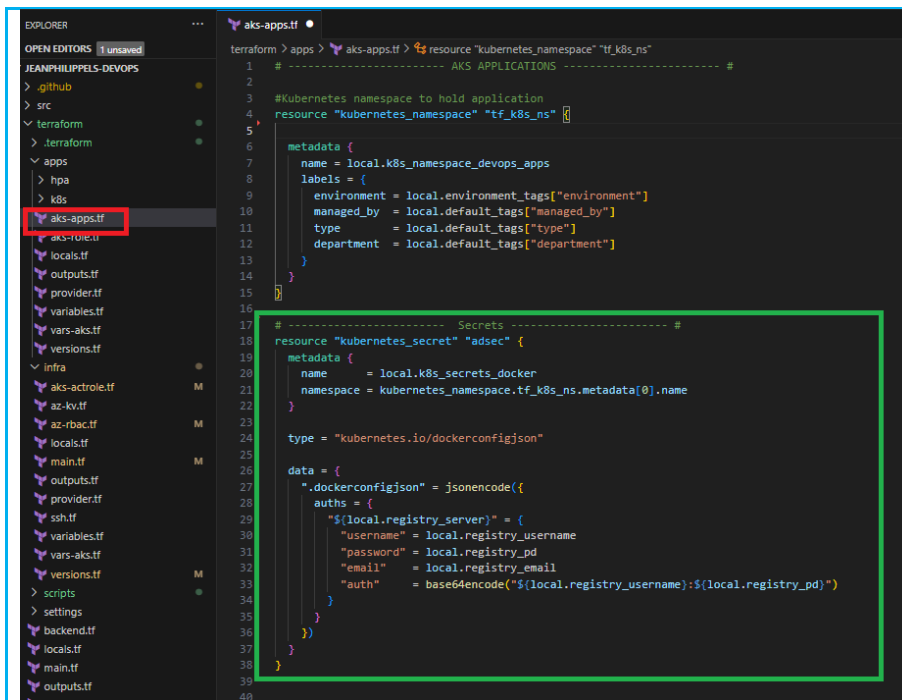
```

Azure Portal (Bottom): The portal shows the 'k8s-dev-cluster | Namespaces' page. The 'Namespaces' tab is selected and highlighted with a red box. The table below lists the namespaces in the cluster:

Name	Status	Age
kube-system	Active	48 minutes
kube-public	Active	48 minutes
kube-node-lease	Active	48 minutes
default	Active	48 minutes
gatekeeper-system	Active	47 minutes
dev-apps-jeanphilipels	Active	44 minutes

The 'dev-apps-jeanphilipels' namespace is highlighted with a green box, indicating it is the namespace created by the Terraform configuration.

○ **Create the Secrets allowing to use private docker repository**



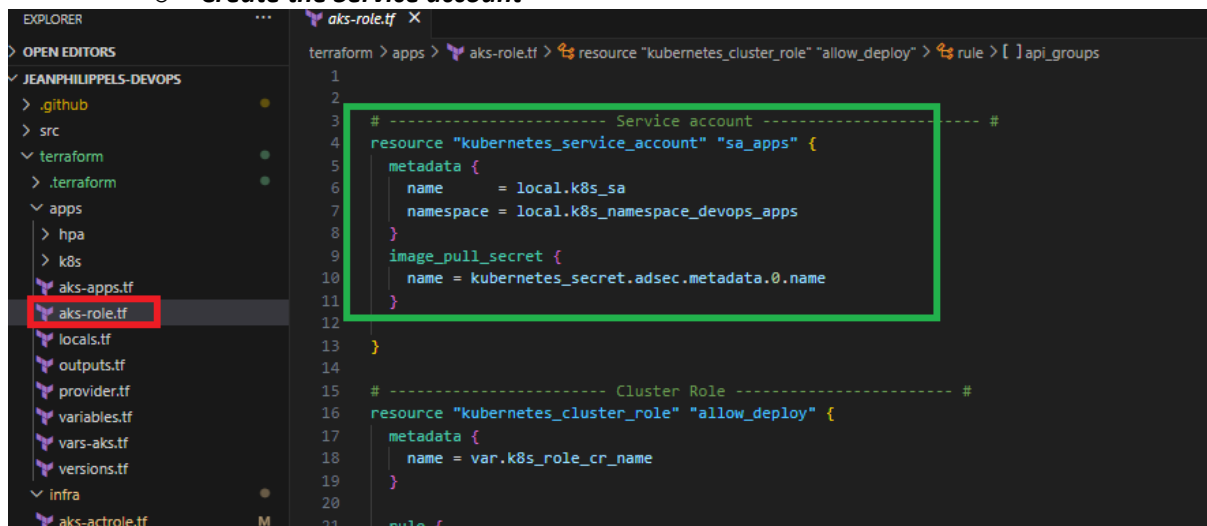
The screenshot shows the Terraform code for `aks-apps.tf`. The left sidebar shows the file explorer with `aks-apps.tf` selected. The main editor displays the following code:

```

1 # ----- AKS APPLICATIONS ----- #
2
3 #Kubernetes namespace to hold application
4 resource "kubernetes_namespace" "tf_k8s_ns" {
5
6   metadata {
7     name = local.k8s_namespace_devops_apps
8     labels = {
9       environment = local.environment_tags["environment"]
10      managed_by   = local.default_tags["managed_by"]
11      type         = local.default_tags["type"]
12      department   = local.default_tags["department"]
13    }
14  }
15 }
16
17 # ----- Secrets ----- #
18 resource "kubernetes_secret" "adsec" {
19   metadata {
20     name       = local.k8s_secrets_docker
21     namespace = kubernetes_namespace.tf_k8s_ns.metadata[0].name
22   }
23   type = "kubernetes.io/dockerconfigjson"
24
25   data = {
26     ".dockerconfigjson" = jsonencode({
27       auths = {
28         "${local.registry_server}" = {
29           "username" = local.registry_username
30           "password" = local.registry_pd
31           "email"    = local.registry_email
32           "auth"     = base64encode("${local.registry_username}:${local.registry_pd}")
33         }
34       }
35     })
36   }
37 }
38
39
40

```

○ **Create the Service account**



The screenshot shows the Terraform code for `aks-role.tf`. The left sidebar shows the file explorer with `aks-role.tf` selected. The main editor displays the following code:

```

1 terraform > apps > aks-role.tf > resource "kubernetes_cluster_role" "allow_deploy" > rule > [ ] api_groups
2
3 # ----- Service account ----- #
4 resource "kubernetes_service_account" "sa_apps" {
5   metadata {
6     name       = local.k8s_sa
7     namespace = local.k8s_namespace_devops_apps
8   }
9   image_pull_secret {
10     name = kubernetes_secret.adsec.metadata[0].name
11   }
12 }
13
14 # ----- Cluster Role ----- #
15 resource "kubernetes_cluster_role" "allow_deploy" {
16   metadata {
17     name = var.k8s_role_cr_name
18   }
19 }
20
21 rule {

```

○ **Create the role cluster for the service account**


```

terraform > apps > aks-role.tf > resource "kubernetes_cluster_role" "allow_deploy" > rule > [ ] api_groups
4 resource "kubernetes_service_account" "sa_apps" {
13 }
14
15 # ----- Cluster Role ----- #
16 resource "kubernetes_cluster_role" "allow_deploy" {
17   metadata {
18     name = var.k8s_role_cr_name
19   }
20
21   rule {
22     api_groups = [""]
23     resources = ["pods"]
24     verbs      = ["list", "get", "watch", "create", "delete"]
25   }
26
27   rule {
28     api_groups = [""]
29     resources = ["pods/exec"]
30     verbs      = ["create"]
31   }
32
33   rule {
34     api_groups = [""]
35     resources = ["pods/log"]
36     verbs      = ["get"]
37   }
38
39   rule {
40     api_groups = [""]
41     resources = ["pods/attach"]
42     verbs      = ["list", "get", "create", "delete", "update"]
43   }
44
45   rule {
46     api_groups = [""]
47     resources = ["secrets"]
48     verbs      = ["list", "get", "create", "delete", "update"]
49   }
50
51   rule {
52     api_groups = [""]
53     resources = ["configmaps"]
54     verbs      = ["list", "get", "create", "delete", "update"]
55   }
56
57   rule {
58     api_groups = [""]
59     resources = ["services"]
60     verbs      = ["list", "get", "create", "delete", "update", "patch"]
61   }
62
63   rule {
64     api_groups = ["apps"]
65     resources = ["deployments"]
66     verbs      = ["list", "get", "create", "delete", "update", "patch"]
67   }
68 }
69

```

○ **Bind the role cluster to the service account**

```

terraform > apps > aks-role.tf > resource "kubernetes_cluster_role_binding" "allow_deploy" > subject > name
16 resource "kubernetes_cluster_role" "allow_deploy" {
51   rule {
52     api_groups = [""]
53     resources = ["services"]
54     verbs      = ["list", "get", "create", "delete", "update", "patch"]
55   }
56
57   rule {
58     api_groups = ["apps"]
59     resources = ["deployments"]
60     verbs      = ["list", "get", "create", "delete", "update", "patch"]
61   }
62 }
63
64 # ----- Cluster Role Binding ----- #
65 resource "kubernetes_cluster_role_binding" "allow_deploy" {
66   metadata {
67     name = local.k8s_role_cr_name_binding
68   }
69
70   role_ref {
71     api_group = "rbac.authorization.k8s.io"
72     kind      = "ClusterRole"
73     name      = kubernetes_cluster_role.allow_deploy.metadata.0.name
74   }
75
76   subject {
77     kind      = "ServiceAccount"
78     name      = kubernetes_service_account.sa_apps.metadata.0.name
79     namespace = kubernetes_namespace.tf_k8s_ns.metadata[0].name
80   }
81 }
82

```

- *Create the custom devops web application deployment with secret from a docker hub repository*

```

77 terraform > apps > aks-apps.tf > resource "kubernetes_deployment" "appsk8s" > spec > template > spec >
78
79 #Kubernetes deployment
80
81 resource "kubernetes_deployment" "appsk8s" {
82   depends_on = [
83     kubernetes_namespace.tf_k8s_ns,
84     kubernetes_service_account.sa_apps,
85     kubernetes_secret.adsec,
86     kubernetes_cluster_role.allow_deploy,
87     kubernetes_cluster_role_binding.allow_deploy, kubernetes_service.appsk8s
88   ]
89   metadata {
90     name      = local.k8s_deploy_name
91     namespace = kubernetes_namespace.tf_k8s_ns.metadata[0].name
92     labels = {
93       app      = "${local.k8s_label_name}"
94       version   = "${var.version_apps}"
95       environment = local.environment_tags["environment"]
96       managed_by  = local.default_tags["managed_by"]
97       type        = local.default_tags["type"]
98       department  = local.default_tags["department"]
99     }
100   }
101   spec {
102     replicas = var.k8s_replicas
103     selector {
104       match_labels = {
105         app      = "${local.k8s_label_name}"
106         version   = "${var.version_apps}"
107         environment = local.environment_tags["environment"]
108         managed_by  = local.default_tags["managed_by"]
109         type        = local.default_tags["type"]
110         department  = local.default_tags["department"]
111       }
112     }
113   }
114   template {
115     metadata {
116       labels = {
117         app      = "${local.k8s_label_name}"
118         version   = "${var.version_apps}"
119         environment = local.environment_tags["environment"]
120         managed_by  = local.default_tags["managed_by"]
121         type        = local.default_tags["type"]
122         department  = local.default_tags["department"]
123       }
124     }
125     spec {
126       service_account_name = local.k8s_sa
127     }
128   }
129 }

```

```

terraform > apps > aks-apps.tf > resource "kubernetes_deployment" "appsk8s" > spec > template > spec
81 resource "kubernetes_deployment" "appsk8s" {
102   spec {
115     template {
127       spec {
129         container {
135           path = "/"
136           port = var.k8s_container_port
137         }
138
139         initial_delay_seconds = 10
140         period_seconds = 5
141         failure_threshold = 3
142       }
143     }
144   }
145
146   liveness_probe {
147     http_get {
148       path = "/"
149       port = var.k8s_container_port
150     }
151   }
152
153   initial_delay_seconds = 10
154   period_seconds = 5
155   failure_threshold = 3
156 }
157
158 resources {
159   limits = {
160     cpu = "20m"
161     memory = "80Mi"
162   }
163   requests = {
164     cpu = "10m"
165     memory = "20Mi"
166   }
167 }
168
169 port {
170   container_port = var.k8s_container_port
171 }
172
173 image_pull_secrets {
174   name = kubernetes_secret.adsec.metadata[0].name
175 }
176
177 }
178

```

- **Create the service custom devops web application deployment**

```

terraform > apps > aks-apps.tf > resource "kubernetes_deployment" "appsk8s" > spec
25 resource "kubernetes_secret" "adsec" {
44 }
45 }
46
47
48
49 #Kubernetes service to access the Apps webpage
50 resource "kubernetes_service" "appsk8s" {
51   /*
52     depends_on = [
53       kubernetes_namespace.tf_k8s_ns, kubernetes_service_account.sa_apps, kubernetes_secret.adsec
54     ]
55     */
56   metadata {
57     name = local.k8s_pods_name
58     namespace = kubernetes_namespace.tf_k8s_ns.metadata[0].name
59   }
60   spec {
61     selector = {
62       app = "${local.k8s_label_name}"
63       version = "${var.version_apps}"
64       environment = local.environment_tags["environment"]
65       managed_by = local.default_tags["managed_by"]
66       type = local.default_tags["type"]
67       department = local.default_tags["department"]
68     }
69     port {
70       port = var.k8s_svc_port
71       target_port = var.k8s_svc_des_port
72     }
73     type = var.k8s_svc_type
74   }
75 }
76
77

```


Plan: 8 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ apps_service_ip = (known after apply)
module.application.kubernetes_namespace.tf_k8s_ns: Creating...
module.application.kubernetes_cluster_role.allow_deploy: Creating...
module.application.kubernetes_cluster_role.allow_deploy: Creation complete after 1s [id=CR-DEV-DEPLOY]
module.application.kubernetes_namespace.tf_k8s_ns: Creation complete after 1s [id=dev-apps-jeanphilippels]
module.application.kubernetes_secret.adsec: Creating...
module.application.kubernetes_service.appsk8s: Creating...
module.application.kubernetes_secret.adsec: Creation complete after 1s [id=dev-apps-jeanphilippels/devops-dev-sec]
module.application.kubernetes_service_account.sa_apps: Creating...
module.application.kubernetes_service_account.sa_apps: Creation complete after 0s [id=dev-apps-jeanphilippels/devopsdevsa]
module.application.kubernetes_cluster_role_binding.allow_deploy: Creating...
module.application.kubernetes_cluster_role_binding.allow_deploy: Creation complete after 1s [id=CR-BIND-dev-DEPLOY]
module.application.kubernetes_service.appsk8s: Creation complete after 9s [id=dev-apps-jeanphilippels/devopsdevpods]
module.application.kubernetes_deployment.appsk8s: Creating...
module.application.kubernetes_deployment.appsk8s: Creation complete after 1s [id=dev-apps-jeanphilippels/devops-deploy-jeanphilippels]
module.application.kubernetes_horizontal_pod_autoscaler.hpaajpls: Creating...
module.application.kubernetes_horizontal_pod_autoscaler.hpaajpls: Creation complete after 1s [id=dev-apps-jeanphilippels/k8s-hpa-dev-jeanphilippels]
```

Warning: Resource targeting is in effect

You are creating a plan with the `-target` option, which means that the result of this plan may not represent all of the changes requested by the current configuration.

The `-target` option is not for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Warning: Applied changes may be incomplete

The plan was created with the `-target` option in effect, so some changes requested in the configuration may have been ignored and the output values may not be fully updated. Run the following command to verify that no other changes are pending:

```
terraform plan
```

Note that the `-target` option is not suitable for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Releasing state lock. This may take a few moments...

Apply complete! Resources: 8 added, 0 changed, 0 destroyed.

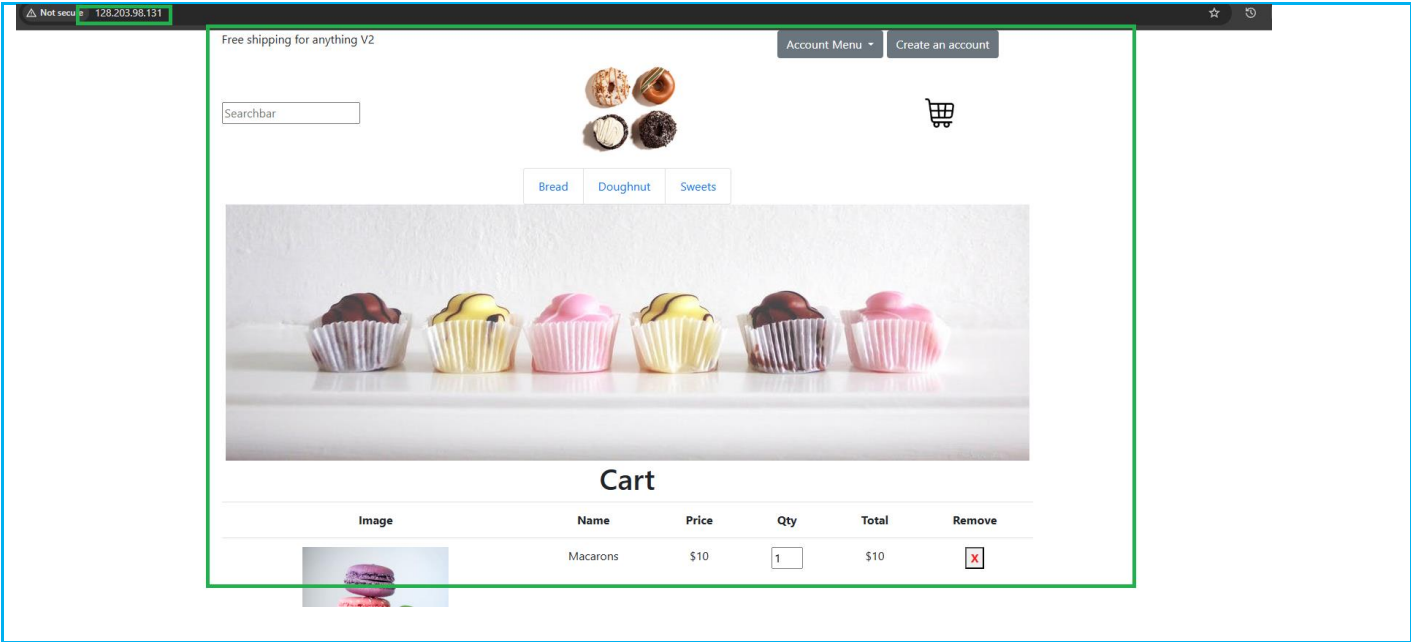
- **Testing the devops web deployment**

```
PS C:\terraform_jpls\Kubernetes_jpls> kubectl get deploy -n dev-apps-jeanphilippels
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
devops-deploy-jeanphilippels        1/1      1              1             4m19s
PS C:\terraform_jpls\Kubernetes_jpls> kubectl get svc -n dev-apps-jeanphilippels
NAME            TYPE           CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
devopsdevpods   LoadBalancer  10.0.121.108  128.203.98.131  80:32748/TCP     29m
PS C:\terraform_jpls\Kubernetes_jpls>
```

```

PS C:\terraform_jpls\Kubernetes_jpls> kubectl describe deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels
Name:          devops-deploy-jeanphilippels
Namespace:     dev-apps-jeanphilippels
CreationTimestamp: Sun, 16 Mar 2025 15:18:35 +1300
Labels:        app=devops_dev_apps
               department=devops
               environment=dev
               managed_by=terraform
               version=V1.0
Annotations:   deployment.kubernetes.io/revision: 1
Selector:      app=devops_dev_apps,department=devops,environment=dev,managed_by=terraform,version=V1.0
Replicas:      1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=devops_dev_apps
          department=devops
          environment=dev
          managed_by=terraform
          version=V1.0
  Service Account: devopsdevsa
  Containers:
    devops-dev-apps:
      Image:   docker1sp123/devops-jeanphilippels:V1.0
      Port:    80/TCP
      Host Port: 0/TCP
      Limits:
        cpu:      20m
        memory:   80Mi
      Requests:
        cpu:      10m
        memory:   20Mi
      Liveness:    http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3
      Readiness:   http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3
      Environment: <none>
      Mounts:       <none>
      Volumes:       <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type            Status  Reason
  ----            -
  Available        True    MinimumReplicasAvailable
  Progressing      True    NewReplicaSetAvailable
OldReplicaSets:   <none>
NewReplicaSet:    devops-deploy-jeanphilippels-8657f6847d (1/1 replicas created)
Events:
  Type    Reason             Age    From                      Message
  ----    -
  Normal  ScalingReplicaSet  4m56s  deployment-controller     Scaled up replica set devops-deploy-jeanphilippels-8657f6847d from 0 to 1
PS C:\terraform_jpls\Kubernetes_jpls>

```



SETTING UP CI/CD PIPELINE

GitHub Actions | Workflow Push Docker image to private repository

- Building DevOps image with Dockerfile
- Building container with the DevOps image
- Testing container with the DevOps image
- Pushing DevOps image to private docker repository

GitHub Actions is a platform that allows to do Continuous Integration and Continuous Delivery (CI/ CD) by automating build, test and deployment in pipeline

• Building DevOps image with Dockerfile

```
github > workflows > tf-docker.yml
1  name: 2-Build and Push docker image
2  description: 'Build and push docker image'
3  on:
4    workflow_run:
5      workflows: ["1-Build and Test"]
6      types: [completed]
7      branches:
8        - 'dev'
9    #workflow_dispatch:
10
11  env:
12    #ENV: "${{ vars.ENV }}"
13    ENV: "dev"
14    version: "V1.0"
15    container_name: "mycaketest-jpls"
16
17  jobs:
18    release-docker:
19      name: Buid | Push Images to docker hub
20      runs-on: ubuntu-latest
21      steps:
22        - name: Check out the repo
23          uses: actions/checkout@v3
24          with:
25            ref: 'dev'
26
27        - name: Build Docker Image
28          run: docker build -t ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }} .
29
```

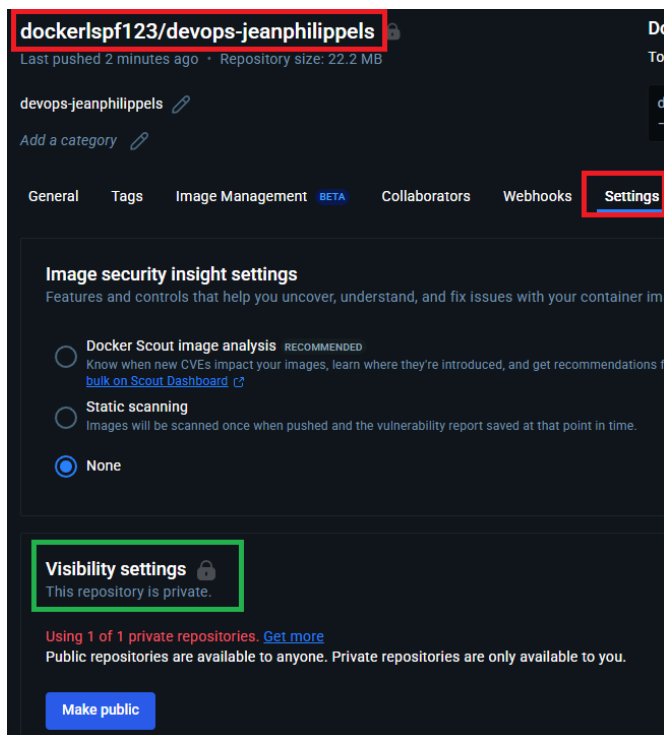
• Building container with the DevOps image

```
29
30  - name: Run Docker Container
31    run: docker run -d -p 80:80 --name $container_name ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}
32
```

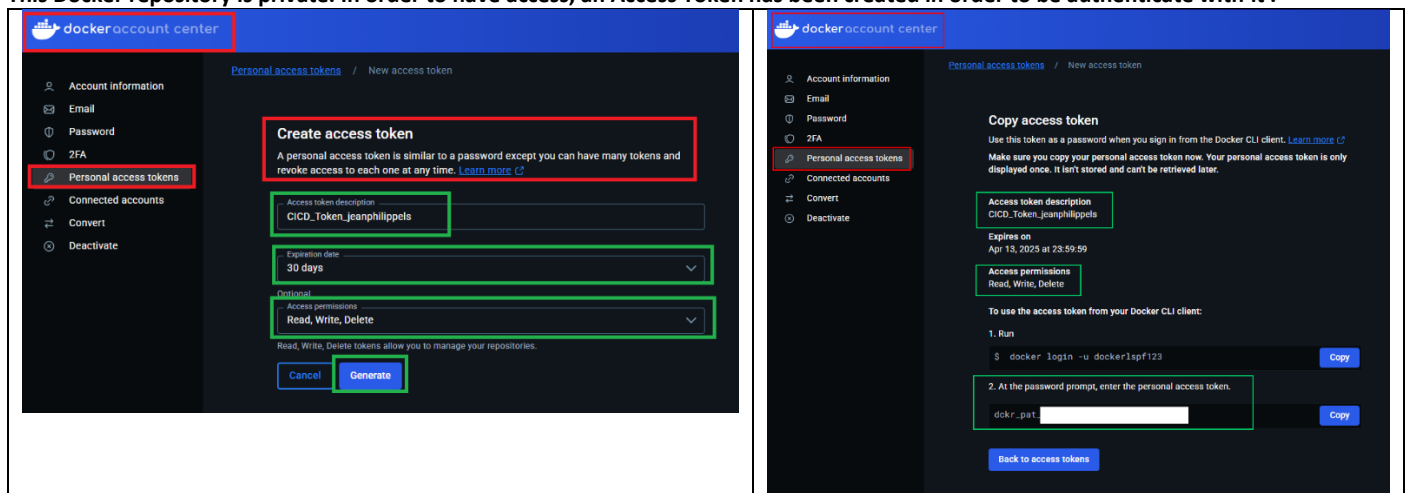
• Testing container with the DevOps image

```
33  - name: test Index Page
34    run: |
35      sleep 10 # Allow container to start
36      curl --fail http://localhost:80/
37
```

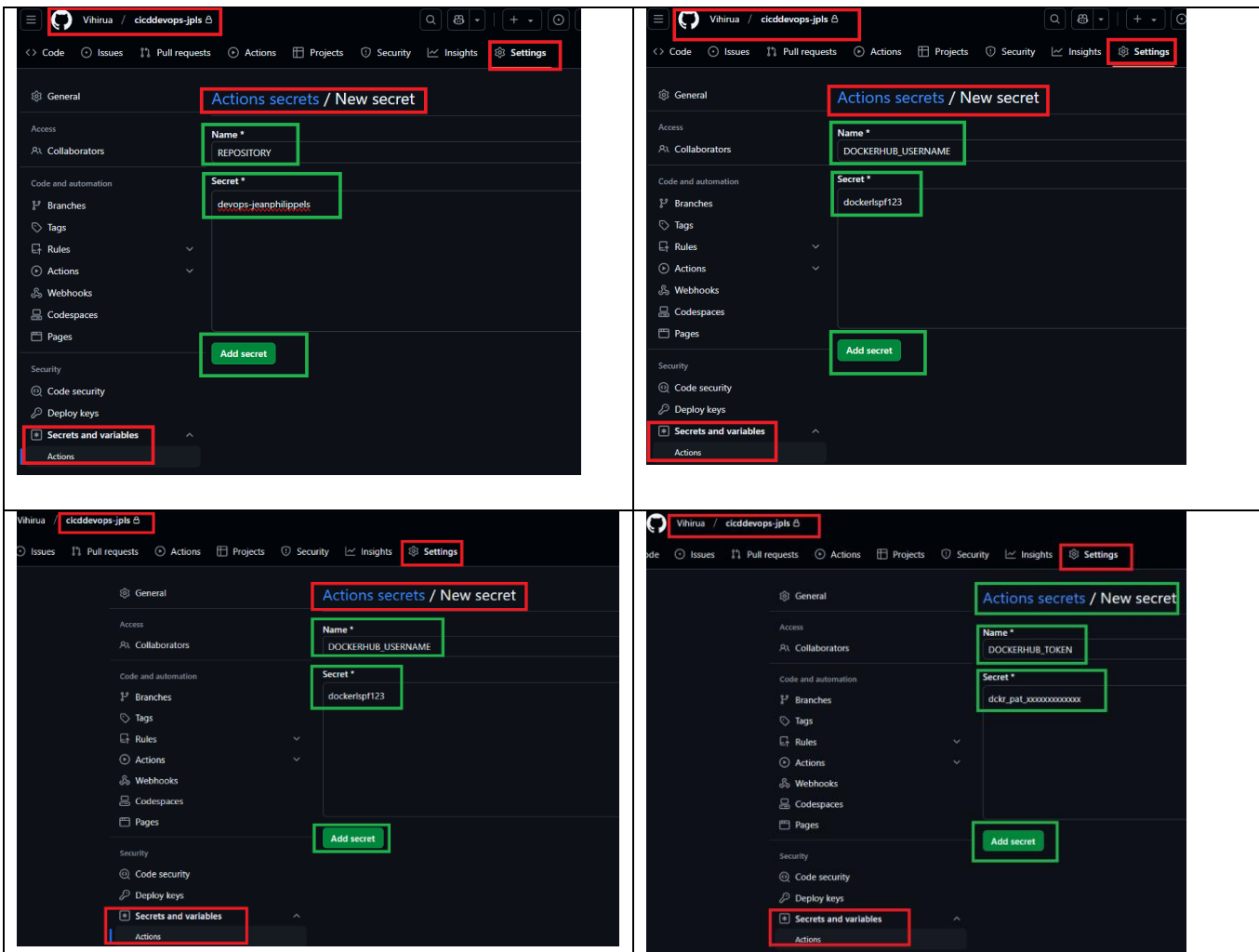

- Pushing images to private docker repository



This Docker repository is private. In order to have access, an Access Token has been created in order to be authenticate with it .



Secrets have been created in GitHub to use for GitHub actions:



Workflow on GitHub for Building and Pushing Docker image into the private docker hub registry :

```

github > workflows > %_ tf-docker.yml
1  name: 2-Build and Push docker image
2  description: 'Build and push docker image'
3  on:
4    workflow_run:
5      workflows: ["1-Build and Test"]
6      types: [completed]
7      branches:
8        - 'dev'
9    #workflow_dispatch:
10
11  env:
12    #ENV: "${{ vars.ENV }}"
13    ENV: "dev"
14    version: "v1.0"
15    container_name: "mycaketest-jpls"
16
17  jobs:
18    release-docker:
19      name: Build | Push Images to docker hub
20      runs-on: ubuntu-latest
21      steps:
22        - name: Check out the repo
23          uses: actions/checkout@v3
24          with:
25            ref: 'dev'
26
27        - name: Build Docker Image
28          run: docker build -t ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }} .
29
30        - name: Run Docker Container
31          run: docker run -d -p 80:80 --name $container_name ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}
32
33        - name: Test Index Page
34          run: |
35            sleep 10 # Allow container to start
36            curl --fail http://localhost:80/
37
38        - name: Stop Test Container
39          run: |
40            sleep 5 # Allow container to stop
41            docker stop $(docker ps -q)
42
43        - name: Delete Test Container
44          run: |
45            sleep 5 # Allow container to be deleted
46            docker rm $(docker ps -a -q)
47
48        - name: Setup Docker
49          uses: docker/setup-buildx-action@v2
50
51        - name: Set Docker Credentials
52          uses: docker/login-action@v2
53          with:
54            username: ${{ secrets.DOCKERHUB_USERNAME }}
55            password: ${{ secrets.DOCKERHUB_TOKEN }}
56
57        - name: Build Image and Push
58          id: build-push-api
59          uses: docker/build-push-action@v3
60          with:
61            context: .
62            file: Dockerfile
63            push: true
64            tags: ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}
65
66

```

A commit into the repository launched the workflow :

The screenshot shows the GitHub Actions interface for the repository 'Vihirua / cicdddevops-jpls'. The 'Actions' tab is selected, showing a workflow run titled '2-Build and Push docker image' with a green checkmark and the number 72. The 'Summary' section on the left lists 'Jobs' and 'Run details'. The 'Jobs' section highlights the job 'Buid | Push Images to docker hub'. The 'Run details' section shows a list of steps, including 'Set up job', 'Check out the repo', 'Build Docker Image', 'Run Docker Container', 'Test Index Page', 'Stop Test Container', 'Delete Test Container', 'Setup Docker', 'Set Docker Credentials', 'Build Image and Push', 'Post Build Image and Push', 'Post Set Docker Credentials', 'Post Setup Docker', 'Post Check out the repo', and 'Complete job'. The job 'Buid | Push Images to docker hub' is highlighted with a green box, and the 'Complete job' step is also highlighted.

After the workflow has been successfully ran, the new image has been push into the private docker repository :

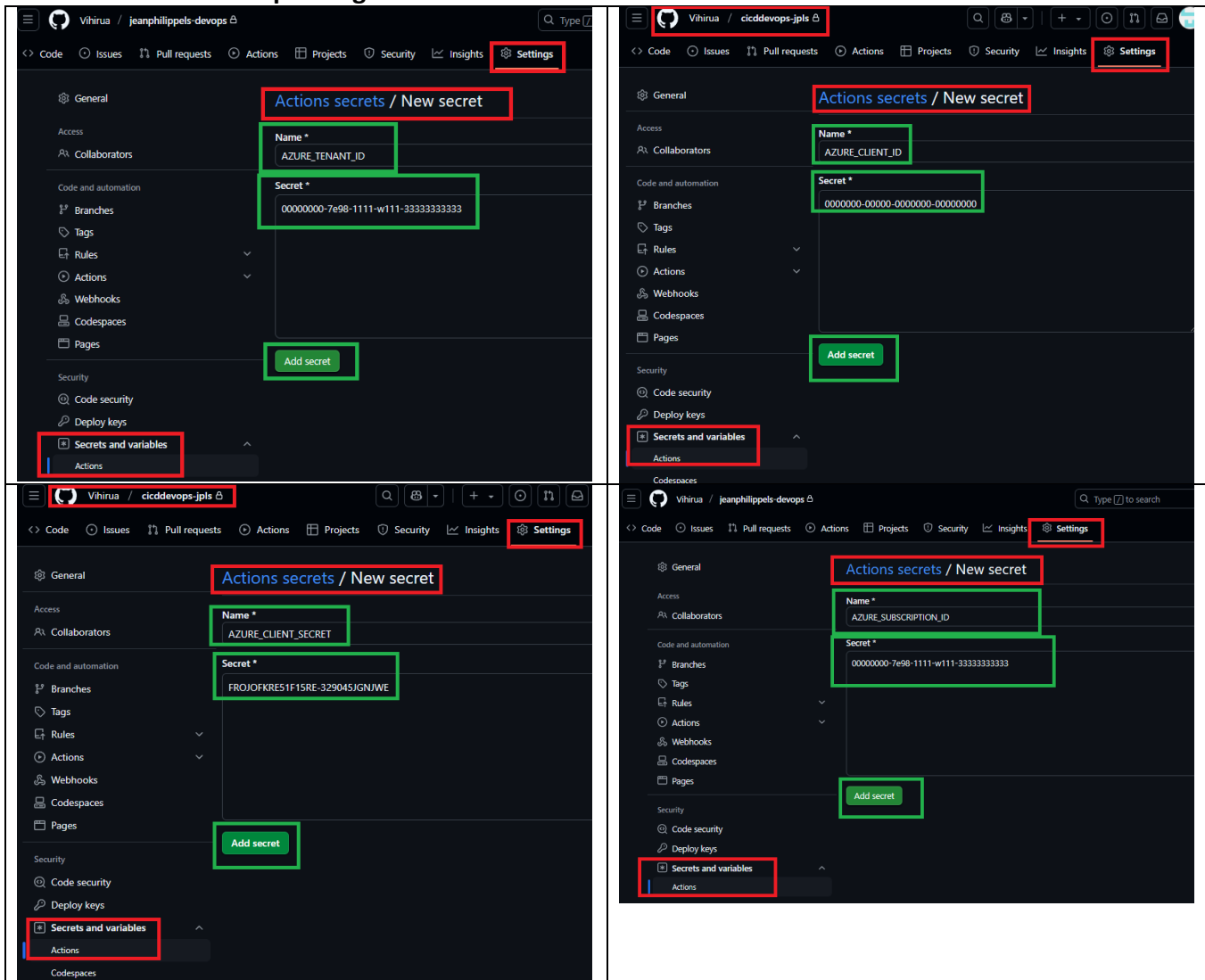
The screenshot shows the Docker Hub repository page for 'dockerlsp123/devops-jeanphilippels'. The repository is highlighted with a red box. The 'Tags' tab is selected, showing a list of tags. The tag 'V1.0' is highlighted with a green box. The 'Tags' section shows the tag 'V1.0' with a green checkmark, last pushed 2 minutes ago by 'dockerlsp123'. The 'Digest' is '6ee15c449e28', the 'OS/ARCH' is 'linux/amd64', the 'Last pull' is 'less than 1 day', and the 'Compressed size' is '20.17 MB'. The 'Docker commands' section shows the command 'docker push dockerlsp123/devops-jeanphilippels:tagname' and the 'docker pull dockerlsp123/devops-jeanphilippels:V1.0' command with a 'Copy' button.

GitHub Actions | Workflow Terraform

Workflow Unit Test

When pushing code to the branch, this workflow will analyze if the scripts are well formed and respond to the format.

- **Test unit when pushing into branches**



```

.github / workflows > ⌘ tf-build_test.yml
1  name: 1-UnitTest
2  description: 'Build and Test code'
3  on:
4    push:
5      branches:
6        - 'dev'
7      paths:
8        - 'terraform/**'
9        - '.github/workflows/tf-build_test.yml'
10 concurrency:
11   group: ${{ github.workflow }}-${{ github.ref }} # Grouping to ensure only one job per branch
12   cancel-in-progress: true
13 env:
14   ARM_CLIENT_ID: ${{ secrets.AZURE_CLIENT_ID }}
15   ARM_CLIENT_SECRET: ${{ secrets.AZURE_CLIENT_SECRET }}
16   ARM_SUBSCRIPTION_ID: ${{ secrets.AZURE_SUBSCRIPTION_ID }}
17   ARM_TENANT_ID: ${{ secrets.AZURE_TENANT_ID }}
18   #ENV: "${{ vars.ENV }}"
19   ENV: "dev"
20   CONF_FILE: "backend-unittest.conf"
21   BACKEND_SCRIPT: "backend-unit-test"
22   BACKEND_RM_SCRIPT: "backend-unit-test-rm"
23   WORKING_DIR: "./terraform"
24   WORKING_SCRIPTS_DIR: "scripts"
25 jobs:
26   terraform-unit-tests:
27     name: "Terraform Unit Tests"
28     runs-on: ubuntu-latest
29     concurrency:
30       group: dev_environment
31       cancel-in-progress: true
32     environment: development
33     defaults:
34       run:
35         shell: bash
36         working-directory: ${{ env.WORKING_DIR }}
37     steps:
38       # Checkout the repository
39       - name: Checkout
40         uses: actions/checkout@v3
41       - name: Initialization of the Backend on Azure
42         id: init-backend
43         shell: bash
44         run: |
45           chmod +x ./${{ env.WORKING_SCRIPTS_DIR }}/${{ env.BACKEND_SCRIPT }}.sh
46           ./${{ env.WORKING_SCRIPTS_DIR }}/${{ env.BACKEND_SCRIPT }}.sh
47       env:
48         AZURE_CLIENT_ID: ${{ secrets.AZURE_CLIENT_ID }}
49         AZURE_CLIENT_SECRET: ${{ secrets.AZURE_CLIENT_SECRET }}
50         AZURE_TENANT_ID: ${{ secrets.AZURE_TENANT_ID }}
51       # Test all Terraform files the latest version

```

```

51 # Install Terraform - the latest version
52 - name: Set up Terraform
53   uses: hashicorp/setup-terraform@v3
54   with:
55     terraform_version: 1.6.0 # Specify Terraform version
56 # Initialize Terraform
57 - name: Terraform Init
58   id: tf-init
59   run: terraform init -reconfigure -backend-config=settings/${ENV}/${CONF_FILE}
60 - name: Destroy Backend on Azure
61   id: destroy-backend
62   shell: bash
63   run: |
64     chmod +x ./${{ env.WORKING_SCRIPTS_DIR }}/${{ env.BACKEND_RM_SCRIPT }}.sh
65     ./${{ env.WORKING_SCRIPTS_DIR }}/${{ env.BACKEND_RM_SCRIPT }}.sh
66   env:
67     AZURE_CLIENT_ID: ${ secrets.AZURE_CLIENT_ID }
68     AZURE_CLIENT_SECRET: ${ secrets.AZURE_CLIENT_SECRET }
69     AZURE_TENANT_ID: ${ secrets.AZURE_TENANT_ID }
70 # Validate terraform files
71 - name: Terraform Validate
72   run: terraform validate
73
74 # Checks that all Terraform configuration files adhere to a canonical format
75 - name: Terraform Format
76   run: terraform fmt -check -recursive
77
78

```

1-Build and Test

tf-build_test.yml

Filter workflow runs



Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback



101 workflow runs

Event

Status

Branch

Actor

✓ Add

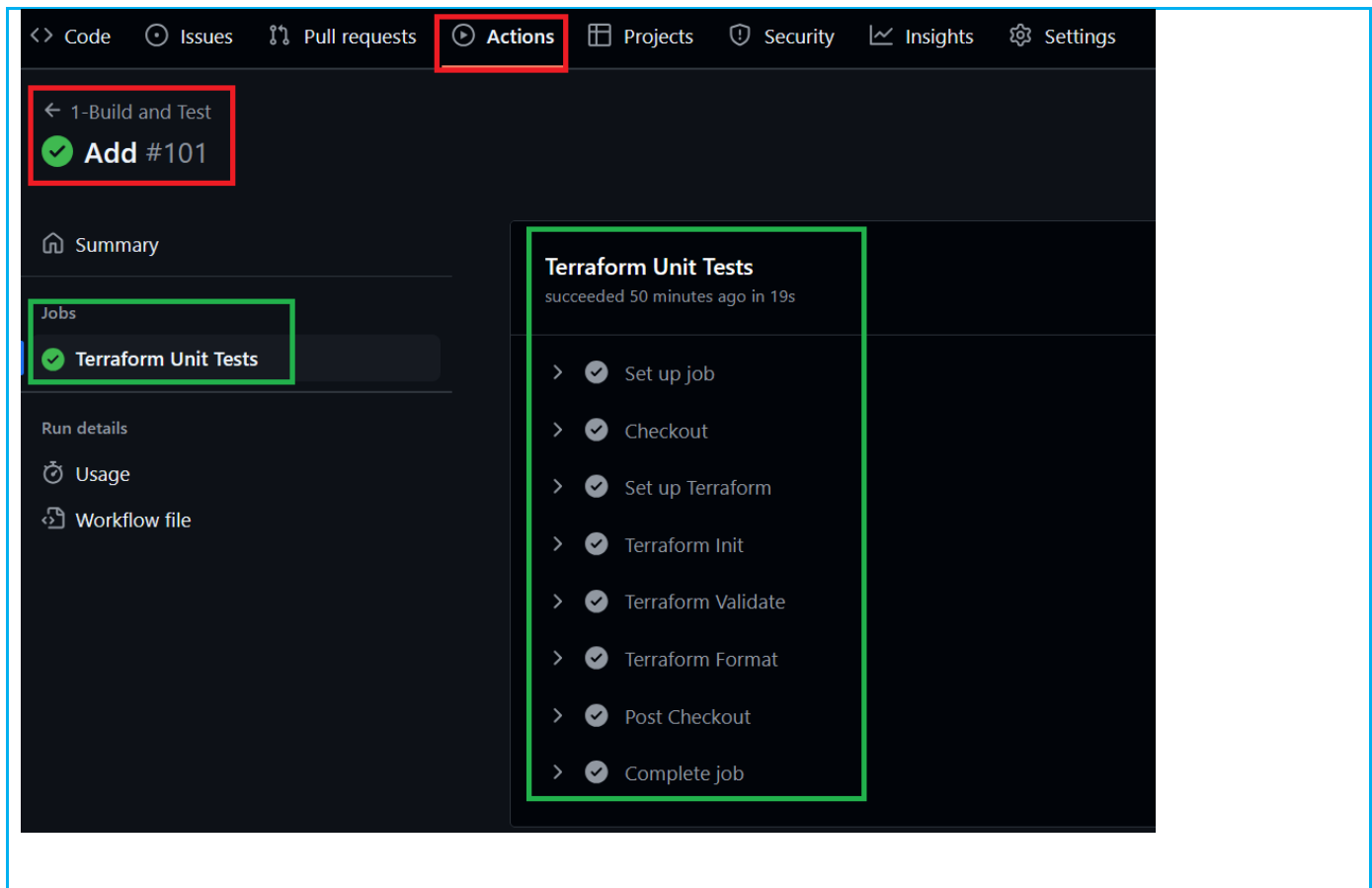
1-Build and Test #101: Commit a8f34b0 pushed by Vihirua

dev

51 minutes ago

25s





Workflow Init Backend

This workflow provisions the backend to use on Azure allowing to test manually module with terraform if teams want to .

- *Shell to create resource group and container for the backend*

The screenshot shows the GitHub Actions interface for a workflow named '9-BackEnd-Init' in the repository 'jeanphilippels-devops'. The 'Actions' tab is selected. The workflow run '#4' is shown as successful. The 'Backend Init' job is highlighted, showing a list of steps: 'Set up job', 'Checkout', 'Initialization of the Backend on Azure' (highlighted with a green box), 'Post Checkout', and 'Complete job'. The 'Initialization of the Backend on Azure' step is marked as successful.

The screenshot shows the Microsoft Azure portal 'Resource groups' page. The search bar contains 'rg-dev-tfstate'. The table below shows the results:

Name ↑	Subscription	Location
rg-dev-tfstate-jpls	Azure for Students	East US

Workflow Azure Kubernetes Services

This workflow provisions Azure Kubernetes Services with the backend and the Azure Kubernetes Services

The image displays two screenshots of the GitHub Actions interface for a repository named 'jeanphilippes-devops'.

Top Screenshot: The 'Actions' tab is selected. The workflow '3-Provision AKS' (file: `tf-publish-todev-nod.yml`) is highlighted. It shows 32 workflow runs. A 'Run workflow' button is visible. A dropdown menu is open, showing 'Use workflow from' with 'Branch: dev' selected and a 'Run workflow' button.

Bottom Screenshot: The '3-Provision AKS #32' workflow run is selected. The 'Summary' tab is active. It shows the workflow was manually triggered 18 hours ago by user 'Vihirua' and is in 'Success' status. The total duration is 7m 28s, and the billable time is 9m. The workflow file is `tf-publish-todev-nod.yml`, triggered by `on: workflow_dispatch`. The workflow steps are: 'Backend Initialization' (1m 1s), 'Terraform Plan' (29s), and 'Terraform Apply'. The 'Re-run all jobs' button is visible.

Summary

Jobs

Backend Initialization

Terraform Plan

Terraform Apply

Run details

Usage

Workflow file

Manually triggered 18 hours ago

Status

Total duration

Billable time

Artifacts

Vihirus

Success

7m 28s

9m

1

tf-publish-todev-nod.yml

on: workflow_dispatch

Backend Initialization

1m 1s

Terraform Plan

29s

Terraform Apply

5m 43s

Terraform Plan summary

Terraform Plan Output

Click to expand

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

<+ read (data resources)

Terraform will perform the following actions:

module.architecture.data.azuread_group.aksgroup will be read during apply

(depends on a resource or a module with changes pending)

<- data "azuread_group" "aksgroup" {

assigned_members = (known after apply)

+ auto_subscribe_new_members = (known after apply)

+ behaviors = (known after apply)

+ description = (known after apply)

Summary

Jobs

- ✔ Backend Initialization
- ✔ Terraform Plan
- ✔ Terraform Apply

Run details

- 🔍 Usage
- 📄 Workflow file

```

# module.architecture.local_file.kubeconfig will be created
+ resource "local_file" "kubeconfig" {
+   content           = (sensitive value)
+   content_base64sha256 = (known after apply)
+   content_base64sha512 = (known after apply)
+   content_md5       = (known after apply)
+   content_sha1      = (known after apply)
+   content_sha256    = (known after apply)
+   content_sha512    = (known after apply)
+   directory_permission = "0777"
+   file_permission    = "0777"
+   filename           = "../kubeconfig"
+   id                  = (known after apply)
}

# module.architecture.random_pet.ssh_key_name will be created
+ resource "random_pet" "ssh_key_name" {
+   id          = (known after apply)
+   length      = 2
+   prefix      = "ssh"
}

Plan: 16 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ aks_msi_principal_id = (known after apply)
+ client_certificate    = (sensitive value)
+ client_key            = (sensitive value)
+ cluster_ca_certificate = (sensitive value)
+ cluster_name          = "k8s-dev-cluster"
+ cluster_username      = (sensitive value)
+ host                  = (sensitive value)
+ kube_config           = (sensitive value)

```

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home >

Resource groups ...

Default Directory (jeanphilippes@gmail.onmicrosoft.com)

+ Create Manage view Refresh Export to CSV Open query Assign tags Group by none

You are viewing a new version of Browse experience. Some features may be missing. [Click here to access the old experience.](#)

rg-dev-k8s-jpls Subscription equals all Location equals all Add filter

Name 1	Subscription	Location
rg-dev-k8s-jpls	Azure for Students	East US
rg-dev-k8s-jpls-nrg	Azure for Students	East US

The image displays two screenshots of the Microsoft Azure portal interface, showing resource groups and their associated resources.

Top Screenshot: Resource group 'rg-dev-k8s-jpls'

- The breadcrumb navigation shows 'Home > Resource groups > rg-dev-k8s-jpls'.
- The 'Overview' tab is selected in the left sidebar.
- The 'Essentials' section shows 'Resources' with a filter for 'k8s-'. The results show 1 record: 'k8s-dev-cluster' (Kubernetes service) located in 'East US'.

Bottom Screenshot: Resource group 'rg-dev-k8s-jpls-nrg'

- The breadcrumb navigation shows 'Home > Resource groups > rg-dev-k8s-jpls-nrg'.
- The 'Overview' tab is selected in the left sidebar.
- The 'Essentials' section shows 'Resources' with a filter for 'Filter for any field...'. The results show 7 records:

Name	Type	Location
416ac108-8245-403b-bbb7-ce34abc7dc80	Public IP address	East US
aks-agentpool-12231162-nsg	Network security group	East US
aks-k8sdevnode-34073874-vmss	Virtual machine scale set	East US
aks-vnet-12231162	Virtual network	East US
azurepolicy-k8s-dev-cluster	Managed Identity	East US
k8s-dev-cluster-agentpool	Managed Identity	East US
kubernetes	Load balancer	East US

Workflow Applications to Azure Kubernetes Services

This workflow provisions the applications into the Azure Kubernetes Services

The screenshot shows the GitHub Actions interface for a workflow named '4-Provision APPS | AKS'. The workflow is in a 'Success' state, triggered manually 18 hours ago. The summary table indicates a total duration of 2m 51s, a billable time of 4m, and 1 artifact. The workflow steps are 'Terraform Plan' (34s) and 'Terraform Apply' (2m 6s). The workflow file is 'tf-publish-todev-apps.yml' on the 'workflow_dispatch' event.

Manually triggered 18 hours ago	Status	Total duration	Billable time	Artifacts
Vihirua -> f8d184f dev	Success	2m 51s	4m	1

tf-publish-todev-apps.yml
on: workflow_dispatch

```

graph LR
    A[Terraform Plan 34s] --> B[Terraform Apply 2m 6s]
  
```

The screenshot shows the 'Terraform Plan Output' section of the workflow run. It displays the Terraform plan summary and the output of the 'Click to expand' section. The output shows that Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols: + create. Terraform will perform the following actions:

```

# module.application.kubernetes_cluster_role.allow_deploy will be created
+ resource "kubernetes_cluster_role" "allow_deploy" {
  + id = (known after apply)
}
  
```

The image displays two screenshots related to a DevOps implementation for provisioning AKS.

Top Screenshot: Terraform Plan Output

The left sidebar shows the navigation menu with 'Summary' selected. The main area displays the Terraform plan output for the 'sa_apps' resource. The plan indicates that 8 resources will be added, 0 will be changed, and 0 will be destroyed.

```
environment = dev
managed_by  = "terraform"
type        = "apps"
version     = "v1.0"

session_affinity = "None"
type             = "LoadBalancer"

port {
  node_port = (known after apply)
  port      = 80
  protocol  = "TCP"
  target_port = "80"
}

# module.application.kubernetes_service_account.sa_apps will be created
resource "kubernetes_service_account" "sa_apps" {
  automount_service_account_token = true
  default_secret_name             = (known after apply)
  id                              = (known after apply)
}

image_pull_secret {
  name = "devops-dev-sec"
}

metadata {
  generation = (known after apply)
  name       = "devopsdevsa"
  namespace  = "dev-apps-jeanphilippels"
  resource_version = (known after apply)
  uid        = (known after apply)
}
```

Plan: 8 to add, 0 to change, 0 to destroy.

Changes to Outputs:

- apps_service_ip - (known after apply)

Bottom Screenshot: GitHub Actions Workflow

The screenshot shows the GitHub Actions interface for the 'jeanphilippels-devops' repository. The 'Actions' tab is selected, showing the workflow '4-Provision APPS | AKS'. The workflow is marked as successful with a green checkmark.

The workflow steps are listed in the 'Terraform Plan' section, which is highlighted with a green box:

- Set up job
- Checkout the repository to the GitHub Action runner
- Set Version of the Apps
- Set up Terraform
- Terraform Init
- Terraform Refresh
- Terraform format
- Terraform plan
- Publish Terraform Plan Applications
- Create String Output
- Publish Terraform Plan to Task Summary
- Post Checkout the repository to the GitHub Action runner
- Complete job

The screenshot displays the GitHub Actions interface for a repository named 'jeanphilippels-devops'. The 'Actions' tab is selected in the top navigation bar. The workflow '4-Provision APPS | AKS' is highlighted in the left sidebar, with its 'Terraform Apply' job also selected. The main panel shows the 'Terraform Apply' job details, indicating it 'succeeded 17 hours ago in 2m 6s'. A list of 18 steps is shown, all of which are completed, as indicated by green checkmarks. The steps include setting up the job, checkout, setting up Terraform, Terraform Init, Terraform Refresh, downloading the Terraform Plan, setting up az login, saving the tf state, running Terraform Apply, getting Terraform Output, showing the output variable, setting up kubectl, setting up kubelogin, setting the AKS context, and finally running 'Get Deploy' and 'Get Services'.

4-Provision APPS | AKS

4-Provision APPS | AKS #47

Summary

Jobs

- Terraform Plan
- Terraform Apply

Run details

Usage

Workflow file

Terraform Apply

succeeded 17 hours ago in 2m 6s

Search logs

- > Set up job
- > Checkout
- > Setup Terraform
- > Terraform Init
- > Terraform Refresh
- > Download Terraform Plan
- > Setup az login
- > Save tf state
- > Terraform Apply
- > Get Terraform Output
- > Showing Output Var
- > Setup kubectl
- > Setup kubelogin
- > Set AKS context
- > Run Get Deploy
- > Run Get Services

Workflow Destroy

This workflow destroys provisioned azure resources

← 10-Terraform Destroy

✓ 10-Terraform Destroy #21

Summary

Jobs

- ✓ Terraform Plan
- ✓ Terraform Apply
- ✓ Backend Destroy

Run details

- Usage
- Workflow file

Manually triggered 19 hours ago

Status: Success

Total duration: 14m 1s

Billable time: 15m

Artifacts: 1

tf-destroy.yml
on: workflow_dispatch

✓ Terraform Plan 3m 40s → ✓ Terraform Apply 8m 38s → ✓ Backend Destroy 1m 27s

Summary

Jobs

- ✓ Terraform Plan
- ✓ Terraform Apply
- ✓ Backend Destroy

Run details

- Usage
- Workflow file

```

      fullnameOverride: "sq-dev-jpls"
      monitoringPasscode: "L92F#503dmkPFNMM"
      postgresql:
        enabled: true
    EOT,
  ] -> null
- verify          = false -> null
- version         = "2025.1.1" -> null
- wait           = true -> null
- wait_for_jobs   = false -> null
}

# module.sec.kubernetes_namespace.terraform_k8s_sq will be destroyed
- resource "kubernetes_namespace" "terraform_k8s_sq" {
  - id              = "sq-dev-jpls" -> null
  - wait_for_default_service_account = false -> null

  - metadata {
    - annotations = {} -> null
    - generation  = 0 -> null
    - labels      = {
      - "department" = "devops"
      - "environment" = "dev"
      - "managed_by" = "terraform"
      - "type"       = "sec"
    } -> null
    - name          = "sq-dev-jpls" -> null
    - resource_version = "4481" -> null
    - uid           = "1641c7d9-4694-48c6-80c4-5f116a6e46b3" -> null
  }
}

Plan: 0 to add, 0 to change, 48 to destroy.
  
```


The screenshot shows the GitHub Actions interface for the workflow '10-Terraform Destroy #21'. The left sidebar contains a 'Jobs' section with three items: 'Terraform Plan', 'Terraform Apply', and 'Backend Destroy', all marked with green checkmarks. The main content area displays the details for the 'Terraform Apply' job, which succeeded 19 hours ago in 8m 38s. The job steps include 'Set up job', 'Checkout', 'Setup Terraform', 'Terraform Init', and 'Download Terraform Plan'. The 'Terraform Apply' step is expanded, showing a list of Terraform commands and their outputs, including the destruction of various Kubernetes resources like 'horizontal_pod_autoscaler', 'helm_release', 'deployment.apps', 'service', and 'cluster_role_binding'.

The screenshot shows the GitHub Actions interface for the workflow '10-Terraform Destroy #21'. The left sidebar contains a 'Jobs' section with three items: 'Terraform Plan', 'Terraform Apply', and 'Backend Destroy', all marked with green checkmarks. The main content area displays the details for the 'Backend Destroy' job, which succeeded 19 hours ago in 1m 27s. The job steps include 'Set up job', 'Checkout', 'Destroy Backend on Azure', 'Post Checkout', and 'Complete job'.